# Children First, Housing Later*

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**Abstract**

Should housing assistance require sobriety? Without externalities, no: conditionality removes an option from forward-looking households. While conditionality reduces the externalities drug use imposes on users' children, it also raises homelessness, an externality that harms children and everyone nearby. A general-equilibrium model of homelessness, drug use, drug prices and mortality, calibrated to United States data, captures both externalities, with altruism matching how much less parents use drugs than non-parents. Full altruism sharply cuts drug use and homelessness. Conditioning assistance raises homelessness but substantially cuts drug use, raising welfare in transition and the long run through children's greater resources and future earnings.

*Keywords:* homelessness; housing vouchers; substance use disorder; intergenerational transmission; externalities; distortions.

*JEL classification:* I18; I38; H23; R21; J62.

## 1 Introduction

On a single night in January 2024 the point-in-time (PiT) count run by the Department of Housing and Urban Development (HUD) found 771,480 people homeless in the United States, the most since the count began. In July 2025 an executive order moved the federal government decisively to one side of a long argument about what to do about it: it required sobriety in federally funded housing and homelessness programmes.[1] The money followed, redirecting about a third of what the programmes at the time carried between them away from Housing First and towards programmes conditional on sobriety.[2] Whether housing assistance should be conditioned on sobriety is no longer an academic question. It is policy that decides the welfare of hundreds of thousands of people.

The argument between the Housing First and Treatment First camps is usually framed as a contest between two behavioural claims.[3] Critics of Housing First say that unconditional assistance is moral hazard: guaranteeing housing whatever a household does weakens the incentive to stay sober and attached to the labour market, so use and substance use disorder (SUD) rise, which feeds back to homelessness, because a SUD costs jobs and a lost job means not only more use but less ability to pay rent. Critics of Treatment First say that conditionality delays stable housing for exactly the households least able to wait, worsens their employment, and makes SUD recovery harder, because the street and shelters are poor places to recover. That too runs in a circle: worse recovery means more disorder, and more disorder means more homelessness. Both claims have merit, and the experimental literature has not settled between them, because the trials that could are short.

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[1] Executive Order, *Ending Crime and Disorder on America's Streets*, 24 July 2025. Reported in `https://www.npr.org/2025/07/24/nx-s1-5479139/trump-homelessness-executive-order-civil-commitment-camping`.

[2] See `https://www.npr.org/2025/11/14/nx-s1-5553561/homelessness-housing-funding-trump-administration-hud`.

[3] For the case against Housing First as the federal government now makes it, see the executive order cited above; for the case for it, see the National Alliance to End Homelessness, `https://endhomelessness.org/blog/ceo-corner-week-of-july-28/`.

Randomised trials of Housing First find large and persistent gains in housing stability and mixed to null effects on substance use (Tsemberis et al., 2004; Aubry et al., 2016, among others), but they focus on the drug users themselves, and run six to twenty four months, so they cannot speak to long-run behaviour or to the budget required to fund either policy at scale. A model can follow what a trial cannot. It can trace the impact of a policy well beyond the households in the study group, to their children and to others in the economy, and well beyond the length of a trial.

This paper starts from the observation that the debate as framed is missing the mark, because the moral hazard both sides argue about has nothing to do with efficiency at all. Consider households that are forward-looking, as in Becker and Murphy (1988), and that have no self-control problems of the kind modelled with hyperbolic discounting at the moment they decide whether to start using. Suppose they account for the true hazard of acquiring a SUD when they make that decision, as the evidence suggests (Lundborg and Lindgren, 2002; van Ours, 1995; Grossman and Chaloupka, 1998; Saffer and Chaloupka, 1999; Caulkins, 2001; Gallet, 2014; Greenwood et al., 2025). Then at the same generosity and the same cost, Housing First weakly dominates Treatment First for every household. Conditionality removes an option from an optimiser and gives nothing back.

The case for Treatment First therefore has to rest on something outside the household's own objective, and this paper focuses on four externalities, which randomised trials cannot resolve: two that Housing First leaves in place and two that Treatment First creates.

The two Housing First leaves in place are harms drug use does to the user's children. The first runs through the household's budget, the *budget externality*: children live with their parents and share their consumption and their dwelling, but not the drug, so every unit a parent spends on drugs, and every job a SUD costs it, is consumption and housing the child goes without. When Treatment First reduces drug use, it diverts more resources towards children. The second runs through earnings in adulthood, the *earnings externality*: a child whose parent had a SUD during its upbringing starts adult life with lower productivity and earns less for the rest of its life. When Treatment First reduces the SUD, it raises the earnings of children in adulthood.

The two Treatment First creates both run through homelessness. By denying housing assistance to drug users, Treatment First exposes more of them to homelessness. The first externality this creates falls on everyone, the *aggregate homelessness externality*: every household's utility falls with the share of the population that is homeless, capturing in a reduced-form way the idea that it is less safe and pleasant to live in a place with more homelessness, and homeless households do not internalise the cost their own homelessness imposes on everyone else. The second externality falls on drug users' children, the *childhood homelessness externality*: a child lives wherever its parent lives, so a parent put in a shelter takes the child into the shelter too.

A parent takes the budget, earnings and childhood homelessness externalities into account only through its altruism, and only in part. How much weight it puts on its child is calibrated to the difference in drug use between parents and adults without children in the data. In the 2024 National Survey on Drug Use and Health (NSDUH), adults without children use drugs 18.3 percent more than parents with the same characteristics, so the weight is set so that removing a parent's concern for its child would raise drug use in the model by that much. The part of the child's loss a parent does internalise is thus pinned down by observed behaviour.

Given the degree of altruism, the size of each externality is set by data. The budget externality depends on how much a user household spends on the drug. The weight given to drugs in the utility function is set to match the share of their income that drug users spend on drugs, 16.2 percent, from retail drug spending (Midgette et al., 2019) and the incomes of adults who use in the 2024 NSDUH. The earnings penalty is set from a direct estimate: children exposed in utero to a Swedish alcohol experiment earn about 24 percent less as adults (Nilsson, 2017). The intergenerational transmission of SUD is then left untargeted: in the model a child of a parent with a SUD is 1.51 times as likely to acquire one itself, against 1.5 to 2.5 in the data (Mellentin et al., 2016; Yoon et al., 2013). The childhood homelessness externality carries no earnings penalty at all. The link between a parent's homelessness and the child's comes from intergenerational transmission alone, and the model is calibrated to estimates of earnings transmission from the literature (Solon, 1992; Mazumder, 2005; Chetty et al., 2014): a child's productivity is correlated with its parent's, and lower-productivity households are poorer and more likely to be homeless, so their children are more

likely to draw a lower productivity, be poor, and end up homeless as well. With this mechanism in place, a child of a homeless parent is 4.92 times as likely to be homeless itself, against 1.8 to 2.4 in the data (Bassuk et al., 1997). The aggregate homelessness externality is priced as a disamenity in flow utility, in proportion to the homelessness rate, set from the contingent valuation in Loubière et al. (2020), where the Europeans willing to pay anything to end homelessness would pay 0.2 percent of average annual earnings a year.

Who bears these externalities depends on who uses drugs, and one mechanism of the model decides it. Two facts about drug use have to be reproduced together, and the literature has so far explained them only by assuming that lower-education households, whose income is lower because of their education, have a stronger preference for drugs (Greenwood et al., 2025). Drug use is concentrated at the bottom of the income distribution, where the average user's income is about 74 percent of the average in the 2024 NSDUH, and it is not universal: most households do not use at all, so there is a margin at zero, with a participation rate of 8.90 percent among adults aged 26 and over in the same survey. To generate both I model drugs as a luxury good bought at a price, which puts most households at a corner where they optimally consume none. Left to its own devices, however, that same assumption concentrates use at the *top*, since the share of income spent on a luxury good rises with income. What reverses the income gradient, as I show, is the increased mortality risk that comes with a SUD: acquiring a disorder truncates the expected continuation value of life, and that loss rises faster than income does, so in equilibrium use concentrates at the bottom.

I proceed with two models, a stylized model for the intuition and a full quantitative model for the policy analysis. The stylized model is a life-cycle model with three periods of life (childhood, adulthood and old age) and rational forward-looking households, who care about consumption, housing and drugs. It shows four things at their clearest: how and why the income gradient in drug use reverses, how the value of life and moral hazard pull drug use in opposite directions, how one can find the optimal degree of homelessness, drug use and SUD, and why the externalities Treatment First removes outweigh the ones it creates.

In the stylized model, an adult draws a productivity and an unemployment shock, and chooses how much of the drug to buy; drug users risk acquiring a SUD, and a SUD raises the chance of dying before old age, the chance of losing employment, and a job lost to the disorder is not covered by unemployment insurance. It chooses whether to live in a shelter or in a rental, and it may hold a means-tested voucher, which is rationed; food assistance puts a floor under its consumption, so that all households have a positive value of life, which means they have something valuable to lose by dying. Children live with their parents during the parents' adulthood and share their resources (except the drugs), so what a parent spends on drugs comes out of its child's consumption and housing, and the weight a parent puts on its child is calibrated, as in the full model, to match the gap in drug use between parents and adults without children. The aggregate homelessness externality is in the model too, calibrated to the same willingness to pay as in the full model. At the end of the parents' adulthood, their child becomes an adult, and its productivity draw is penalised if the parent acquired a SUD while the child lived with them. There is no government budget in the stylized model, so revoked vouchers are not offset by a higher probability of drawing a voucher for sober households; that is deliberate, so the mechanisms can be seen clearly, and the quantitative model takes up the detail. The model is calibrated to United States data.

Welfare is measured using the ex-ante expected utility of newborns, childhood included, in consumption-equivalent terms, against the world with rationed Housing First.

This stylized model shows starkly how the income gradient reverses. It separates the two penalties a disorder carries (death and unemployment). Scaled from nothing to 2 times its calibrated value, with the other penalty switched off, the death penalty takes the average user from 3.54 times average income to 0.52 times it; the unemployment penalty on its own leaves it at 3.54, against 3.54 with neither penalty. The reason is how each penalty varies with income. A fixed quantity of the drug costs a rich household less in utility terms than a poor household, which means that the utility price of using falls as income rises. A job the disorder costs drops a household to the same floor no matter what it earned before, which leaves its resources equal to its food assistance, its savings and, under Housing First, a voucher if it draws one. Food assistance and the free shelter put the same floor under every household, which, with risk aversion above one, bounds the utility loss from losing employment, while the utility cost of the drug declines with income, so the unemployment penalty cannot reverse the income gradient in drug use. In contrast, mortality destroys the whole continuation value, including the value of being alive at all, which is worth more to richer households, so when the death penalty is large enough it is the poor who use and the rich who do not.

The experiment is the reform itself. The programme stays at its benchmark rationing, with the same probability of drawing a voucher, and only the rule changes, from Housing First to Treatment First, which does not provide vouchers to households that use drugs or have a SUD. The stylized model keeps the price of drugs fixed; in the quantitative model it is endogenous.

Comparing each rule with no programme at all, and the two with each other, separates the two forces housing assistance sets against each other. Under Treatment First there is no moral hazard, so only the value of life is at stake: assistance makes life worth more, and thus makes the risk of dying from a SUD costlier. Under Housing First moral hazard operates on top of the value of life, and thus the gap in drug use between the two isolates the moral hazard.

The first result of the stylized model is the optimum. To estimate the optimal degree of drug use and homelessness, I set the weight a parent puts on its child to one, so that nothing a parent does to its child is left external. In that economy nobody uses drugs, so nobody acquires a SUD, and homelessness drops by 4 percent.

The second result is that, without the externalities, the stylized model confirms the dominance result: 9.4 percent of households are strictly worse off under Treatment First and none is better off. However, counting what drug use costs the child, both the resources a parent diverts towards drugs and the earnings penalty of a parent's SUD, turns the ranking, from the first cohort of newborns on. This is because in this model the deterrence is large enough that no household holding a voucher chooses to use in the long run, and a household deterred by Treatment First from using never acquires the disorder, keeps more of its resources for its children, and its children inherit better draws. In the stylized model Treatment First lowers childhood homelessness as well, by about 1 percent, because fewer households have a SUD and lose their jobs with no income support, and fewer are poor for having been born to a parent with the disorder; the quantitative model does not share this last result regarding childhood homelessness. In the transition, the old and the adults alive at the reform lose 0.07 and 0.017 percent of consumption, while their children, the first cohort born into the reform, already gain 0.36 percent of lifetime consumption, because their parents are deterred while they live with them.

The full quantitative model that follows quantifies in more detail the mechanisms and the welfare implications of Housing First against Treatment First, and settles whether that ranking survives in a richer environment that fits the data. It is a general-equilibrium life-cycle economy, and households here, as in the stylized model, care about consumption, housing and drugs. Children live with their parents and share consumption and housing resources with them while the parents are 26 to 50 years old, and become adults after that. Households choose whether to live in a rental or a shelter, and whether to use drugs. Using carries the risk of a SUD, from which a household recovers only with a small probability each year: recovery is exogenous, harder in a shelter and with serious mental illness (SMI), but more likely in treatment, which a household can choose at a cost. SMI arrives exogenously with age and is more likely for households with a SUD, and like a SUD it lowers employment and survival, all consistent with the data (Milby et al., 2005; Olfson et al., 2015; Nielsen et al., 2017; Murrie et al., 2020; Hjorthøj et al., 2023).

The price of drugs clears the drug market, so drug prices respond to policy, and in general equilibrium, as drug prices decline, using drugs becomes more appealing, which is a force pulling against the deterrence effect of Treatment First. The government budget balances through the probability of receiving a voucher, so the programme's reach responds to what it costs. Conditionality is enforced by a drug test that the government pays for but that misses some users; a household that is caught loses its voucher and is barred from the programme for a time. The model is calibrated to United States data as well, with a benchmark of rationed Housing First, in which about a quarter of eligible households get a voucher with no sobriety conditions.

The model is also validated on a long list of moments that no parameter is fitted to, set out in Section 5.3. Among the larger ones: the homelessness rate by age band and the age composition of the homeless, both from the HUD PiT count; the share who return to homelessness within two years; drug poisonings as a share of deaths between 25 and 44; the earnings held by the bottom and top tenths and the income and net worth held at the very bottom of the distribution; the share of renters spending more than thirty and more than fifty percent of income on housing; the labour force participation of households with and without a disorder, and the income of those with one; how often the homeless enter treatment, and what they bring to it; the price elasticity of drug participation; and the odds that the child of a parent with a SUD acquires one, and

that a homeless adult had a homeless caretaker. Each is a place the model can be caught.

The quantitative experiments start from full altruism, which shows how much is at stake: if parents weighed their children fully, drug use would fall by 36 percent and homelessness by 13 percent, even under rationed Housing First. The Treatment First reform itself is the same rationed change of rule as in the stylized model, with the difference that the probability of drawing a voucher rises to balance the budget. In the long run Treatment First cuts drug use by 30 percent and the SUD by 31 percent, but raises homelessness by 6 percent. The externalities it removes outweigh the ones it creates: a newborn gains 1.39 percent of lifetime consumption, 50 percent of it during childhood, from its parent spending more on its consumption and housing instead of on drugs, and the rest as an adult, only about 21 percent of that through the higher earnings of a child whose parent is free of the disorder, even as childhood homelessness rises by 6 percent, from 1.02 to 1.08 percent of children. An earnings penalty for growing up in a shelter would not reverse this result: the share of children in shelters rises by only 0.06 percentage points, while the share of children living with a parent who has a SUD falls by 1.59 points, from 5.28 to 3.69 percent.

Accounting for the transition, the reform lands on an economy that already has many households with a SUD. Since recovery from a SUD happens only with a small probability, for many their SUD lasts a while, and their demand keeps the drug price elevated for many years. They are the first to lose their vouchers and be exposed to higher risk of homelessness, but the elevated price deters young adults who still have children living with them. In the first decade the number of drug users without a SUD falls by up to 48 percent and homelessness rises by up to 14 percent, against 29 and 6 percent in the long run, respectively. Children born at the reform gain about 2.1 percent of lifetime consumption, against 1.39 percent in the long run, because their parents are the young adults the elevated price keeps from starting. That drop in drug use, larger in the transition than in the long run, diverts more resources to them while they live with their parents, so their gain during childhood is 2.1 times the long-run gain, despite a small rise in their homelessness (as vouchers are revoked for some households who use drugs), by less than in the long run, 0.04 percentage points, from 1.02 to 1.06 percent.

## 2 Related literature

The literatures this paper draws on are set out in the order the argument needs them. The first three supply the machinery: what the trials establish about housing policy, how economists model the decision to use substances, and what structural models of homelessness already deliver. The next two concern the externalities, which is where this paper departs from its predecessors: what a parent with a SUD transmits to their child, and how much weight a parent puts on the child who bears it. The last two are what the quantitative model is built out of: the hazards, taken one at a time from clinical and epidemiological work, and the drug market that makes the price endogenous.

**Randomised trials of Housing First.** A large empirical literature evaluates programmes that provide rapid housing without conditioning on sobriety, and finds large and persistent improvements in housing stability and reductions in shelter and other public service use (Tsemberis et al., 2004; Padgett et al., 2006; Larimer et al., 2009; Goering et al., 2014; Aubry et al., 2016). Evidence on substance use is mixed: most studies find no statistically significant change, and some report modest reductions in alcohol related outcomes. The trial this paper draws on directly is Milby et al. (2005): among cocaine-dependent homeless adults, all of them in day treatment, those randomised into housing stayed abstinent longer than those who were not, and it is where the model takes the penalty homelessness imposes on recovery from.

However, randomised trials typically measure outcomes over six to twenty four months and rarely beyond three years, which is short against a recovery process that Vaillant (1995) documents as much more gradual and protracted: at the annual recovery rate the model takes from it, 22 percent of those with a SUD recover within ten years. A further limitation of a trial, and perhaps a deeper one, is its scope. A trial randomises households within a fixed environment, and holds fixed exactly the things that decide the welfare comparison: the price of the drug, the budget that pays for a programme at scale, and, most importantly, the effects on children and other households that are not in the sample.

**Economic models of drug use.** Modelling substance use as forward-looking optimisation dates to Becker and Murphy (1988), and the assumption is empirically defensible at the margin where policy operates. Both users and non-users hold risk perceptions close to the objective probabilities (Lundborg and Lindgren, 2002), and Greenwood et al. (2025), who allow perceived and true SUD risk to differ, find that the model-implied beliefs are close to the true risk. A separate literature emphasises temptation and dynamic inconsistency (Laibson, 1997; Gruber and Kőszegi, 2001), under which present-biased households over-consume relative to their own long-run preferences and corrective policy has a rationale with no externality at all. That literature is complementary here rather than opposed, and it cuts in the same direction: a present-biased household would welcome a sobriety requirement as a commitment device, so introducing it would strengthen the case for conditionality rather than weaken it.

Where this paper departs materially from that literature is in how it generates the income gradient in drug use. Greenwood et al. (2025) build a macroeconomic model of the opioid crisis in which households differ by skill, and the concentration of use among the less skilled comes about because of preference differences towards drugs across those groups. In contrast, in the model here no preference parameter varies with income: every household has the same weight on the drug, the same Stone-Geary endowment, the same hazard of acquiring a SUD and the same mortality penalty. The utility function does not change with the state the household is in, or with the choices it makes. The gradient comes out of the mortality channel alone, because a SUD destroys a continuation value and the continuation value rises with permanent income, so households with more to lose lose proportionally more than they gain from a drug that is cheap relative to their income.

**Structural models of homelessness.** The paper builds directly on İmrohoroğlu and Zhao (2022), who model homelessness as a rational response to income risk, mental health shocks, housing costs and housing policy. In that framework housing policy reaches homelessness through income loss and housing costs alone, so a sobriety-contingent policy has no behavioural margin to act on in their model. Adding endogenous use, disorder and recovery, together with state-dependent mortality, creates that margin: the same voucher that insures a household also changes what using costs it, and the disorder it may acquire then feeds back into employment, housing and survival. Abramson (2026) makes the complementary point from the landlord side of the market, showing that eviction policy changes landlord screening and rents and thus changes who becomes homeless in equilibrium.

In this model the causation between the disorder and homelessness runs both ways, and the two directions reinforce each other. On the one hand, a SUD costs employment, which costs a household its housing. On the other hand, losing housing worsens employment outcomes and raises mortality, which raises the value of using relative to the value of life being risked, since households with less continuation left to lose risk less by using drugs. Each direction feeds the other, so what the model produces is a vicious circle rather than a one-directional flow.

**Intergenerational transmissions.** Three links have to hold for the argument to work, and each is estimated separately.

The first is that a parent's SUD changes the childhood. The administrative record of the child welfare system documents it at scale: parental substance abuse is recorded at more than a third of removals. The epidemiological literature documents it too. Household substance abuse is one of the adverse childhood experiences of Felitti et al. (1998), and children of a parent with a SUD are between one and a half and two and a half times more likely to acquire one themselves when one parent is affected (Mellentin et al., 2016; Yoon et al., 2013).

The second link is that the childhood shows up in the child's own adult earnings, and the mechanism is as follows. A parent who has lost their job due to their SUD, and who spends what money is left on the drug, has less of it for the child. Time investment in the child goes the same way. What the child loses is schooling, a good peer group, and the ordinary attention a household gives, in a home more chaotic than it would otherwise be. The child reaches the labour market with less human capital and fewer good connections.

Each step of that has been measured. Dahl and Lochner (2012) find that family income raises children's achievement, using the Earned Income Tax Credit to move the income, and Restuccia and Urrutia (2004)

put roughly half of the intergenerational correlation in earnings down to parental investment in education. Card (1999) puts the return to a year of schooling at around ten percent. And Nilsson (2017) closes the loop on a parental substance shock in particular: children exposed in utero to a Swedish experiment that sharply raised alcohol availability earn about twenty four percent less as adults and finish less schooling. On the transmission of the disorder itself, Mellentin et al. (2016) follow 398,881 offspring in the Danish registers and Yoon et al. (2013) a United States population-based sample, and both find that a child with one affected parent is more likely to acquire a SUD, at odds ratios of 1.44 and 2.51, respectively.

What this paper adds is not another estimate but a price: it puts these intergenerational effects inside an equilibrium model where the parent's use decision is endogenous, so the transmitted loss becomes a quantity that policy moves rather than a coefficient.

The third is that homelessness is persistent across generations as well. Earnings are persistent across generations: estimates of the intergenerational elasticity of earnings in the United States run from about four tenths (Solon, 1992) to about six tenths (Mazumder, 2005), and Chetty et al. (2014) find the same persistence in ranks across the whole distribution. The model carries this persistence from parent to child, and that channel alone is enough to explain, inside the model, the higher likelihood observed in the data (Bassuk et al., 1997) that children of homeless parents are homeless themselves: a child of a homeless parent tends to draw a lower productivity because its parent's was low, not because of the homelessness itself. There is evidence that a poor environment lowers a child's earnings beyond what the persistence of skills across generations explains, although it concerns poor neighbourhoods rather than homelessness. Chyn (2018) finds that children displaced by Chicago's public housing demolitions earned sixteen percent more a year as adults. Chetty et al. (2016) estimate that a move to a lower-poverty neighbourhood before age thirteen raises adult earnings by about a third, and the gain falls with the age at the move, which says that what is transmitted is the environment a child is raised in rather than anything the child carries.

The model imposes no earnings penalty for growing up in a shelter, but adding one would not change its conclusions: under Treatment First the share of children in shelters rises by only 0.06 percentage points, while the share of children living with a parent who has a SUD falls by 1.59 points. Bassuk et al. (1997) measure how much more likely a child of a homeless parent is to be homeless itself. Its case-control design is what makes it usable: among 220 homeless mothers and 216 low-income housed mothers, a mother whose female caretaker had ever been homeless has 1.80 times the odds of being homeless herself, and one whose male caretaker had, 2.41 times.

**How much a parent weighs its child.** The extent to which externalities exist depends on the degree to which the parent does not internalise them, that is, one minus the weight it puts on its child, and that weight has a literature of its own. Altonji et al. (1997) test pure altruism directly, on the transfers parents make to their grown children, and reject the idea that parents are perfectly altruistic: moving a dollar of income from a child to its parents raises what they transfer to it by less than thirteen cents, against the full dollar complete altruism requires. Parents are probably altruistic, but not completely, which is the case for treating that weight as strictly between zero and one rather than assuming either end of it.

The Altonji et al. experiment cannot be run here. It needs a parent and a grown child living as separate households, and in the model the two never overlap: the child lives in its parent's household only while the parent is 26 to 50 years old, and the model does not track the relationship between the two after the child becomes an adult. What the model can do is another test, and for this paper it is a more relevant one: it compares parents with adults without children. In NSDUH 2024 past-month use among parents is 0.845 of use among non-parents, reweighted to the non-parents' joint distribution of age, income, sex and marital status so that the gap is behaviour rather than composition or observed characteristics. The weight on the child's utility in the household's problem is the only thing in the model that makes a household use less than it otherwise would, so the weight is set so that setting it to zero (mimicking having no children) raises drug use by the same 18.3 percent.[4] The externality is therefore not assumed to be large; its size is calibrated to observed behaviour on exactly the margin this paper is about.

[4]Drug users are nearly as likely as anyone to have children: in the National Survey of Family Growth 2017–2019, 75 percent of men and women aged 40 to 49 who used cocaine, crack or methamphetamine or injected a drug in the past year have had a child, against 81 percent of those who used none.

**What the hazards are built from.** The model's transitions are not free parameters dressed as structure; each is anchored on, or checked against, a clinical or epidemiological estimate, and the calibration section discusses them one at a time. Four groups do the work. On mortality, the excess death rate of a SUD is calibrated inside the model, to the average income of users, since it is what decides how far down the income distribution drug use sits. The rate it lands on, 4.00 times the baseline, is in line with the estimates in the literature: Plana-Ripoll et al. (2019) put the mortality rate ratio of a SUD at 3.91 in the Danish registers, and Heiberg et al. (2018) at 6.6 among Norwegians in specialist care for one. Meyer et al. (2023) put the death rate of homeless adults at 1.60 times that of the housed at every age, and Olfson et al. (2015) measure the excess death rate for serious mental illness (SMI); against those hazards the value of a life comes from the willingness-to-pay estimates surveyed in Viscusi and Aldy (2003).

On the mental illness chain, Nielsen et al. (2017) identify the effect of a SUD on the onset of SMI in Danish registers, Gutwinski et al. (2021) meta-analyse its prevalence among the homeless, and Bradizza et al. (2006) review relapse under dual diagnosis.

On recovery, Vaillant (1995) documents how slow unassisted remission is, and Milby et al. (2005) supply a causal estimate: a randomised trial on cocaine-dependent homeless adults in which the housed arm was abstinent 6.60 weeks against 4.48 in the unhoused arm, which is where the penalty homelessness imposes on recovery comes from.

On employment and the cost of housing loss, Desmond and Gershenson (2016) estimate what losing housing does to holding a job, a penalty the model also applies to finding one, and Loubière et al. (2020) price the disamenity of visible homelessness through a contingent valuation, which is what makes the aggregate headcount enter anybody's utility at all.

**The drug market.** Making the drug price endogenous is what lets a policy that changes the number of users change what using costs everybody else. Two elasticities govern it and they do different jobs, so the literatures behind them are worth separating: the demand side is driven by how much the extensive and the intensive margins respond to a given price, and the supply side determines how much the price moves when demand changes.

*The price elasticity of demand.* The demand side has two margins and the model produces both. The first is the extensive margin (participation), which is whether a household uses at all. And the second is the intensive margin, how much a user buys once using. Neither is calibrated. Both are produced by the model and checked against the literature in Section 5.3.

Both margins have been measured empirically, and between them the estimates bound how much a policy can reduce use once the drug price falls: if both elasticities are low, a decline in use that lowers the price does not bring many users back. Estimates of the participation elasticity run from about $-0.3$ to $-0.9$ (Saffer and Chaloupka, 1999; Grossman and Chaloupka, 1998), and Grossman and Chaloupka (1998) find that a permanent ten percent fall in the price of cocaine raises the number of cocaine users by a little over eight percent. The model's participation elasticity, $-2.00$, with the distribution of households held fixed, is higher than these estimates, so in the model a fall in the drug price brings back more users than it would in the data. That works against Treatment First: while it deters use, the drop in price brings back more users than the evidence suggests, so the model if anything understates how much the policy reduces drug use, and the welfare gain that comes from lower use.

The intensive margin is the smaller of the two, between about $-0.3$ and $-0.8$. That same ten percent fall raises the frequency of use among those users by a little over three percent, and Olmstead et al. (2015) put the conditional elasticity for heroin at $-0.8$, on two research designs that share a sample of regular users but not an identification strategy. The model's intensive elasticity, measured over all users, $-0.45$, is in the middle of these estimates, so the model matches how much each user's purchases respond to the drug price.

A separate point about rational drug use follows from the findings of van Ours (1995), Saffer and Chaloupka (1999) and Caulkins (2001): participation responds to price, even for drugs as habit-forming as heroin and cocaine, and long-run responses are consistently larger than short-run ones. That ordering is the signature of forward-looking behaviour rather than myopia, which is why it belongs with Becker and Murphy (1988) above as well as here.

*The elasticity of supply.* This is the general equilibrium offset, and it runs against Treatment First in

particular: deterring households from using takes them out of the market, which causes demand and the drug price to fall, but then a cheaper drug draws others in. I am not aware of a direct estimate for an illicit drug. Cannabis is the closest market I found to an illicit drug, as it remains a Schedule I substance federally, so even a licensed producer works without interstate trade, without ordinary banking and under federal prohibition, which is the set of frictions an illicit supplier faces. Mace et al. (2020) estimate a price elasticity of supply of 1.46 on Washington State administrative data, and that is the elasticity the model reproduces, which is at the low end of what the literature uses: where studies of illicit markets take a position, they treat supply as very elastic, because the drugs are cheap and simple to produce (Rhodes et al., 2002).

**In a nutshell.** The trials establish that housing stability responds to policy but cannot price what happens in the long run, to others not in the sample, or to the budget that pays for it. The rational addiction literature models the use decision but abstracts from housing, children, and mental illness. The structural homelessness literature delivers a housing market equilibrium but has no drug margin for conditionality to act on. The transmission literature measures what having a parent with a SUD does to a child but takes the parent's behaviour as given. This paper's contribution is to put them in one framework and then to insist on a distinction none of them needs on its own: that the moral hazard which dominates the policy debate is not a reason for conditionality, and that whatever case exists has to be made on the externalities.

# 3 A stylized model

## 3.1 Three questions

In the data, drug use is concentrated among low-income households, most of the homeless do not carry a SUD, and housing vouchers reach only about a quarter of the households eligible for them. The quantitative model is built to match facts of that kind. It has seven state variables and fifty five parameters, which is what the welfare and fiscal questions require, but it also makes it hard to see which pieces of the environment do the heavy lifting. This section strips the environment down to where the main questions of this paper are still well posed, to help see the main drivers in the model.

Households are perfectly forward-looking, here and in the quantitative model. They know the true hazard of acquiring a SUD, they are aware of what carrying one would cost them, and they make no mistakes. Nothing below rests on misperception, myopia, self control, or on a taste that changes once the household has developed a SUD. The model abstracts from government and from how the programmes are funded; both enter the quantitative model of Section 4. The stylized model answers three main questions.

1. **Who uses drugs?** In the data, use is concentrated at the bottom of the income distribution, and it is not universal there either: most poor households do not use at all. Both facts have to hold together. I model drugs as a *luxury* good bought at a price, which is what delivers the second: not all households choose to consume drugs. However, left to itself the Stone-Geary form in equation (2) below means the share of income spent on drugs rises with income, so when the environment is calibrated to the use rate, use is concentrated at the *top*. Two risks push the other way, and each costs the household something: a SUD raises the chance of losing work, which costs earnings, and it raises the chance of dying, which costs the value of the life it has left. The question is which of the two reverses the concentration from the top to the bottom, and I show that it is the risk of death.
2. **How does housing assistance shape the drug use decision?** Housing assistance pulls drug use in two opposite directions. It raises the value of staying clean, because a housed future is worth more than a homeless one and using puts that future at risk: I refer to it as the value-of-life force, and it lowers use. In contrast, if the voucher is unconditional, then it cushions the consequence of acquiring a SUD, because a household that acquires one is housed anyway: I call this the moral-hazard force, and it raises use. Both are first order, so theory does not sign the effect of unconditional housing assistance on drug use, and the question is quantitative.
3. **When is Treatment First welfare improving relative to Housing First?** In this model, without externalities, never. Housing First always raises welfare when agents are forward-looking, as they are

in the models in this paper. A SUD creates a genuine tension, since the same voucher that insures a household also makes the risk it insures more attractive, but that tension can only lower welfare because Treatment First only removes an option that a household who does not have hyperbolic discounting can commit to taking under Housing First as well. However, if parents do not fully weigh their children's welfare in their decisions, then drug use and SUD generate externalities, such as the budget, earnings and childhood homelessness externalities, and Treatment First may be preferred. By setting the weight parents give to their children's utility in their own objective function, the model can also say what the optimal drug use, SUD and homelessness rates are.

The model is built in stages, and each is recalibrated before it is used, so every result below is read off a model that fits its targets. The first stage has no child living with its parents and no externalities: nothing a household does lands on anyone else. It answers the first two questions and shows that Treatment First cannot raise welfare. The second adds the child, who lives in the household during the parent's adulthood period, shares its consumption and housing, and inherits a lower lifetime productivity if the parent acquired a SUD in that period. This stage brings in the budget, the earnings and the childhood homelessness externalities, shows what drug use would be if parents counted their children fully, and compares Treatment First with Housing First, in the long run and along the transition to it. The third adds the aggregate homelessness externality, the disamenity every household suffers due to the homelessness of others around them.

The model is qualitative throughout: the calibration is rough by design, there is no validation exercise, and components that obscure the mechanisms have been removed with the model refitted without them. What remains are the main drivers of the results. The welfare evaluations below are for illustration and intuition only; the full quantitative model is what settles the main question, which is whether Treatment First raises welfare relative to Housing First in a model that fits the data.

## 3.2 Stage 1: households on their own

### 3.2.1 Households and preferences

There are two adult periods (adulthood and old age) and a continuum of households. A period stands for thirty years of adult life, so the two of them span ages twenty six to eighty five. Parameters with a natural annual counterpart are compounded to that length: the discount factor is $\beta$ per period and $\bar{\phi}$ is survival over thirty years. Saving earns no interest, so $R = 1$ and a unit saved is a unit carried into the next period. Notation follows the full model in Section 4.

A household is characterised by its productivity $y$, fixed over its life and lognormally distributed across households, $\log y \sim \mathcal{N}(-\frac{1}{2}\sigma_y^2, \sigma_y^2)$, so mean income is one. Labour is supplied inelastically. Households differ in $y$, in age $j \in \{1, 2\}$, in employment status $e \in \{0, 1, 2\}$, in voucher receipt $\eta \in \{0, 1\}$ and in SUD status $d \in \{0, 1\}$, all discussed below, and in nothing else.

Flow utility does not count the child, absent in this version of the model. Non-durable consumption $c$, housing services $h$ and drug consumption $v$ form the bundle

$$X = \left[c^{1-\gamma} h^{\gamma}\right]^{1-\theta} (b + v)^{\theta}, \tag{1}$$

and flow utility is

$$u(c, h, v) \;=\; U(X) \;+\; \aleph, \qquad U(X) = \frac{X^{1-\sigma}}{1-\sigma}. \tag{2}$$

The weight on housing $\gamma$ is its share of the non-drug bundle, the weight on the drug $\theta$ is its share of the whole, and $\sigma$ is the coefficient of relative risk aversion. For simplicity, drugs can be chosen only in the first period, and housing is chosen each period from a three-point grid $\{h_1, h_2, h_3\}$. The pair $(\theta, b)$ governs the drug term: the marginal utility of the $v$th unit is proportional to $\theta/(b + v)$, so $\theta$ mostly sets how much a user uses and $b$ whether anyone uses at all. The Stone-Geary form keeps the first unit's marginal utility finite, which is what generates non-users at all, because without the shifter every household uses whatever the consequences. Finally $\aleph$ is the flow value of being alive, the utility of surviving into a period over and above what is consumed in it, which gives mortality a cost that does not vanish even when consumption is low. Households save in one risk-free asset and cannot borrow.

### 3.2.2 Timing

**Period 1.** Every household enters the period clean. Employment $e \in \{0, 1\}$ is drawn with the unemployment probability $u_0$, and a household that is income eligible draws a voucher with probability $\pi_V$; these are the same two lotteries it faces again in period 2. It then chooses consumption, housing, drug use $v \geq 0$ and saving, paying market rent $p_r$ on whatever unit it picks. It starts with no assets.

Using drugs carries the risk of a SUD, the same whatever the quantity,

$$\pi(v) \;=\; \pi_d \, \mathbf{1}\{v > 0\}. \tag{3}$$

The whole risk therefore sits on the decision to use at all, and $\theta$ alone sets how much a user uses once it has decided to actually use.

**Acquiring a SUD and the death penalty.** The SUD $d' \in \{0, 1\}$ is realised with probability $\pi(v)$ for households that used and with probability zero for the rest, and the household survives into period 2 with probability

$$\phi(0) \;=\; \bar{\phi}, \qquad \phi(1) \;=\; \bar{\phi}^{\,\kappa_{\phi d}}, \tag{4}$$

because a SUD multiplies the annual hazard of dying by $\kappa_{\phi d}$, and a hazard multiplied by $\kappa_{\phi d}$ in every year of a thirty-year period raises survival over the period to the power $\kappa_{\phi d}$. The mortality penalty applies here, and here only.

**Period 2.** This is the terminal period, after which the household dies with certainty. A household without a SUD is unemployed with probability $u_0$ and one with a SUD with the higher probability $u_1$; if unemployed it receives income support, unless the disorder is what cost it the job (Section 3.2.3). The voucher is drawn again, subject to the policy rule (Section 3.5). The household chooses consumption and housing, there is no drug use decision, and assets are run down with nothing bequeathed.

### 3.2.3 Employment, income support and housing

A SUD costs a household resources in one way only: it raises the chance of losing its job from $u_0$ to $u_1$, and income support does not cover the extra job losses. A household with a SUD loses its job for the ordinary reasons anyone does, with probability $u_0$, and because of the disorder with the further probability $u_1 - u_0$. Employment status $e$ therefore takes three values: working ($e = 1$), out of work and covered by income support ($e = 0$), and out of work because of the disorder and not covered ($e = -1$). Income in a period, given permanent productivity $y$ and employment status $e$, is

$$y^g(y, e) \;=\; \begin{cases} y & \text{if } e = 1, \\ \psi_{ui} \cdot \min\{y, \bar{y}\} & \text{if } e = 0, \\ 0 & \text{if } e = -1, \end{cases} \tag{5}$$

with $\psi_{ui}$ the replacement rate and $\bar{y}$ a cap on the earnings base, and $e = -1$ occurs with probability $u_1 - u_0$ for a household with a SUD and zero for every other. There is no other income risk.

Food assistance is paid in kind. A household with income $y^g$ in a period receives

$$F(y^g) \;=\; \max\{0,\; \bar{c} - \rho_F \, y^g\} \tag{6}$$

of consumption, which it cannot spend on rent or on drugs, so its other income has to cover those. I include it because with it every household has a positive value of life: it puts a floor under the consumption of every household, including one that loses its job to a SUD, so that every household has something valuable to lose by dying.

Each period the household chooses

$$h_k \in \{h_1,\; h_2,\; h_3\}, \qquad h_1 < h_2 < h_3, \tag{7}$$

a shelter ($h_1$), a small unit ($h_2$) and a standard unit ($h_3$). A shelter is free and delivers less housing services than the market does.

A voucher covers part of the rent, is means-tested and is rationed. It is drawn independently each period, which is a simplification relative to the quantitative model, where a voucher persists as long as eligibility holds. Writing $\eta \in \{0, 1\}$ for whether one is held,

$$\Pr(\eta' = 1 \mid y', d', v) \;=\; \pi_V \cdot \mathbf{1}\{y' \le \kappa_V \, y^{\text{med}}\} \cdot \mathbf{1}\{\text{policy covers } (d', v)\}, \tag{8}$$

where $\kappa_V \, y^{\text{med}}$ is the threshold for eligibility, $y^{\text{med}}$ is median income, and a household holding one pays the fraction $\kappa_r$ of market rent, so the rent paid on housing option $k$ is

$$r_k(\eta) \;=\; p_r \, h_k \big[\kappa_r \eta + (1 - \eta)\big] \cdot \mathbf{1}\{k \neq 1\}, \tag{9}$$

which is zero when the household is homeless. Vouchers operate in both periods, and $\pi_V$, the share of eligible households actually covered, is a policy dial of Section 3.5.

The two policies differ in exactly one respect, the last indicator in equation (8). **Housing First** (HF) covers every eligible household regardless of $(d', v)$. **Treatment First** (TF) covers only households that satisfy the sobriety condition, $v = 0$ and $d' = 0$, which is the rule in the quantitative model as well. In the stylized model, detection of drug use and SUD is assumed to be perfect, but I relax this assumption in the full model. The condition is applied to the period in which the voucher is paid, so what it tests differs by period: every household is born clean, so $d' = 0$ holds for everyone in period 1 and only $v$ can fail the condition in that period, while in period 2 there is no drug choice, $v = 0$ holds by construction, and only the disorder, $d' = 1$, can fail it. Since policy coverage in the states that follow drug use is the only difference, the gap between them is the moral hazard.

The sobriety condition binds in period 1 as well, and the timing there is worth stating. Every household arrives clean, so the period 1 voucher is drawn before $v$ is chosen and conditionality cannot screen anyone out on entry. The order within the period is therefore: the household chooses $v$, the policy rule is then applied to the voucher it drew, and it chooses housing knowing its voucher outcomes and the effective rent it faces. So under Treatment First a household that uses (period 1), or carries the disorder (period 2), is subject to full market rent immediately, in the period concerned.

### 3.2.4 The household's problem

Let $a$ be assets carried in. Period 2 is terminal, so the household consumes everything and chooses housing,

$$V_2(a', y, e', \eta', d') = \max_{\{k\}} \Big\{ u(c, h_k, 0) \Big\}, \qquad c = Ra' + y^g(y, e') - r_k(\eta') \;+\; F\big(y^g(y, e')\big) \;\ge\; F\big(y^g(y, e')\big), \tag{10}$$

where consumption cannot fall below the food assistance, which is paid in kind. I write $V_2(a', y, d') = \mathbb{E}\big[V_2(a', y, e', \eta', d') \mid y, d'\big]$ for its expectation over the period 2 employment and voucher draws, the SUD having already been realised at the transition. In period 1 the household draws $(e, \eta)$ and then chooses housing, drug use and saving,

$$V_1(y, e, \eta) = \max_{\{k,\, v \ge 0,\, a' \ge 0\}} \Big\{ u(c, h_k, v) + \beta \, \mathbb{E}_{d'} \Big[ \phi(d') \, V_2(a', y, d') \Big] \Big\}, \tag{11}$$

with $c = y^g(y, e) - r_k(\eta) - p_v v - a' \;+\; F\big(y^g(y, e)\big) \;\ge\; F\big(y^g(y, e)\big)$, and $d' = 1$ with the probability of (3). Survival sits inside the expectation over $d'$ because it depends on which $d'$ is realised. A household uses if the value at its best positive $v$ exceeds the value at $v = 0$.

## 3.3 Calibration

Every target below is documented in full in the replication package's data appendix, `data/DATA_APPENDIX.md`, which names the dataset behind each one, its year and universe, and the computation that turns it into the number quoted here. What follows gives the moment and its source; the appendix gives the arithmetic.

Each stage is fitted to the same targets. Table 1 lists what is set outside the model and Table 2 what is fitted inside it, with a column for each stage: the baseline of this section, and the model with the child that Section 3.7 adds, which is refitted with the child in place. The fit uses the quantitative model's rule, moving all internal parameters together, each in proportion to the gap on its own moment until each gap is closed.

### 3.3.1 External calibration

Preferences and demographics are conventional: the discount factor over a thirty-year period is $\beta = 0.86$, which is 0.995 a year compounded,[5] risk aversion is $\sigma = 2$, as in the quantitative model, and survival over a period is $\bar{\phi} = 0.91$, from the Social Security Administration (SSA) period life table for twenty six to fifty five.

The chance that use causes a SUD, $\pi_d$, can also be externally calibrated. This is because drug use and the SUD are fully separated by cohort: the young are the users, and the old carry the SUD but cannot start using. So the SUD rate is participation times $\pi_d$, and $\pi_d$ is the ratio of the two in NSDUH 2024, 3.68 percent of adults aged 26 and over carrying a moderate or severe SUD against 8.90 percent using, which gives $\pi_d = 0.41$.

Income support replaces half of previous earnings on a base capped at mean income, in line with statutory state rules, and nothing after a job lost to the disorder, which those rules treat as misconduct (U.S. Department of Labor, 2024). Food assistance tapers at $\rho_F = 0.30$, the benefit reduction rate of the Supplemental Nutrition Assistance Program (SNAP). The rent per unit of housing services $p_r$ and the price of a unit of drugs $p_v$ are normalised to one and held fixed in every counterfactual. Both rental units are set, the small one at $h_2 = 0.10$ and the standard one at $h_3 = 0.40$ of mean income. The voucher parameters are institutional: covered tenants pay 30 percent of market rent, eligibility is capped at half of median income, $\kappa_V = 0.50$, and $\pi_V = 0.27$ is the ratio of recipients to the income-eligible population in the 2023 Picture of Subsidized Households (PSH), published by HUD.

### 3.3.2 Internal calibration

The dispersion of log permanent income $\sigma_y$ is set to match how unequal earnings are at the bottom: in the 2022 Survey of Consumer Finances (SCF), households aged 26 to 65 with positive wage and salary income hold 17.2 percent of total earnings in the poorer half of the distribution. The weight on housing $\gamma$ is picked to match the 26.8 percent of non-drug expenditure that shelter plus utilities take in the 2024 Consumer Expenditure Survey (CEX). The flow value of being alive $\aleph$ is matched to a value of a statistical life (VSL) of 80 times average annual income, about \$9.0 million against a mean household income of \$113,000, which is within the range surveyed by Viscusi and Aldy (2003) once their estimates are put in current dollars. The pull of the first unit $b$ is chosen to match a drug participation rate of 8.90 percent from NSDUH 2024 for respondents aged 26 and over with a disorder of any severity, and the weight on drugs $\theta$ to aggregate drug spending over the aggregate income of users, 16.0 percent, from retail spending in Midgette et al. (2019) and the incomes of users in NSDUH 2024. The level of food assistance $\bar{c}$ is fitted so that total food assistance is 0.42 percent of total labour income, the ratio of SNAP outlays to gross domestic product (GDP) in the data.

The mortality rate ratio $\kappa_{\phi d}$ is set to match the income of users, 0.74 of the population average in NSDUH 2024, because the death risk is what decides how far down the income distribution use sits, as Section 3.4 shows. Housing services in a shelter $h_1$ are chosen so that the homelessness rate in the model is 0.60 percent, inside a band of 0.40 to 1.00 percent of households. The floor is HUD's 2023 PiT count, a snapshot of a single night; the ceiling is the share homeless at some point over a year, from Burt et al. (2001). The unemployment probability without the disorder $u_0$ is chosen such that the aggregate unemployment rate is 4.85 percent, the average over 2014 to 2024 in the Bureau of Labor Statistics (BLS) Current Population Survey (CPS), and the unemployment probability with the disorder $u_1$ such that the share of the homeless

[5] The annual discount factor here is higher than the one in the quantitative model, and this is deliberate. A period here is thirty years, so every consequence of using drugs arrives thirty years after the decision, whereas in the quantitative model consequences arrive annually and compound. A low annual discount factor would weigh that distant death much less and the mortality channel would be losing to impatience rather than to anything the model is about, so a higher factor corrects for the coarseness of the horizon.

Table 1: Externally set parameters

| Parameter | Description | Source or rationale | Value |
|---|---|---|---|
| *Preferences and demographics* | | | |
| $\beta$ | Discount factor over a thirty-year period | 0.995 annually, compounded | 0.86 |
| $R$ | Gross return on saving | Zero percent interest | 1.00 |
| $\sigma$ | Risk aversion | Standard; Attanasio and Weber (2010) | 2 |
| $\bar{\phi}$ | Survival over a period, no SUD | SSA period life table, ages 26–55 | 0.91 |
| *The SUD* | | | |
| $\pi_d$ | Chance that use starts a SUD | NSDUH 2024: SUD rate over participation | 0.41 |
| *Income* | | | |
| $\psi_{ui}$ | Income support replacement rate | DOL, statutory UI rates | 0.50 |
| $\bar{y}$ | Cap on the earnings base | DOL, state benefit maxima | 1.00 |
| $\rho_F$ | Food assistance taper | SNAP benefit reduction rate | 0.30 |
| *Housing, drugs and the voucher* | | | |
| $p_r$ | Rent per unit of housing services | Normalisation | 1.00 |
| $p_v$ | Price of a unit of drugs | Normalisation | 1.00 |
| $h_2$ | Small rental unit | Externally set | 0.10 |
| $h_3$ | Standard rental unit | Externally set | 0.40 |
| $\kappa_r$ | Tenant's share of the rent | HUD voucher rules | 0.30 |
| $\kappa_V$ | Voucher means test | HUD income limits | 0.50 |
| $\pi_V$ | Chance of drawing a voucher | HUD 2023, recipients over eligible | 0.27 |
| *The child, stage 2 onward* | | | |
| $\xi$ | Equivalence scale while the child is at home | Square-root scale | 1.41 |
| $\rho$ | Persistence of productivity across generations | Solon (1992); Mazumder (2005) | 0.50 |
| $\delta_d$ | Child's productivity penalty, parent had a SUD | Nilsson (2017), a 24 percent cut | 0.27 |

Notes: stage 1 is the model of Section 3.2, without the child; the parameters of the child enter at stage 2, in Section 3.7.

Table 2: Internally calibrated parameters, by stage

| | | | Stage 1 | | Stage 2 | |
|---|---|---|---|---|---|---|
| Parameter | Target moment | Data | Model | Value | Model | Value |
| $\sigma_y$ | Bottom half's share of earnings, SCF 2022 | 17.2% | 21.1% | 0.99 | 21.1% | 0.99 |
| $\gamma$ | Housing over non-drug expenditure, CEX 2024 | 26.8% | 26.8% | 0.40 | 26.8% | 0.40 |
| $\aleph$ | VSL over income | 80 | 84.0 | 11.23 | 80.0 | 13.31 |
| $b$ | Drug participation, NSDUH 2024 | 8.9% | 8.9% | 0.054 | 8.9% | 0.053 |
| $\theta$ | Drug spending over users' income | 16.0% | 15.8% | 0.18 | 16.1% | 0.18 |
| $\kappa_{\phi d}$ | Users' income over the average, NSDUH 2024 | 0.74 | 0.75 | 2.23 | 0.74 | 2.32 |
| $h_1$ | Homelessness rate | 0.6% | 0.6% | 0.041 | 0.6% | 0.042 |
| $u_0$ | Unemployment rate, BLS CPS 2014–24 | 4.9% | 4.8% | 0.047 | 4.8% | 0.047 |
| $u_1$ | SUD share of the homeless, Gutwinski et al. (2021) | 21.7% | 21.8% | 0.100 | 22.1% | 0.100 |
| $\bar{c}$ | Food stamp spending over labour income | 0.42% | 0.42% | 0.11 | 0.42% | 0.11 |
| $\omega_c$ | Use of non-parents over parents, less one, NSDUH 2024 | 18.3% | – | – | 18.2% | 0.0021 |

Notes: stage 1 is the model of Section 3.2, without the child; stage 2 adds the child of Section 3.7 and refits every parameter; $\omega_c$ exists only once the child does.

carrying a SUD is 21.7 percent, from the meta-analysis of Gutwinski et al. (2021). That share rises with $u_1$: every job the disorder costs is one that income support does not cover, so a household with a SUD that has too little saved to pay the rent through it ends up in a shelter.

## 3.4 Who uses drugs

This section asks which feature of the model puts drug use at the bottom of the income distribution, and it starts from the force that any answer has to overcome. Drugs are a luxury good bought at a price, so a household with twice the income gives up less utility per dollar spent on them. With no consequences at all for acquiring a SUD that is the only force operating, and drug use concentrates at the top: in the baseline of stage 1 the average user then earns 3.54 times the population average.

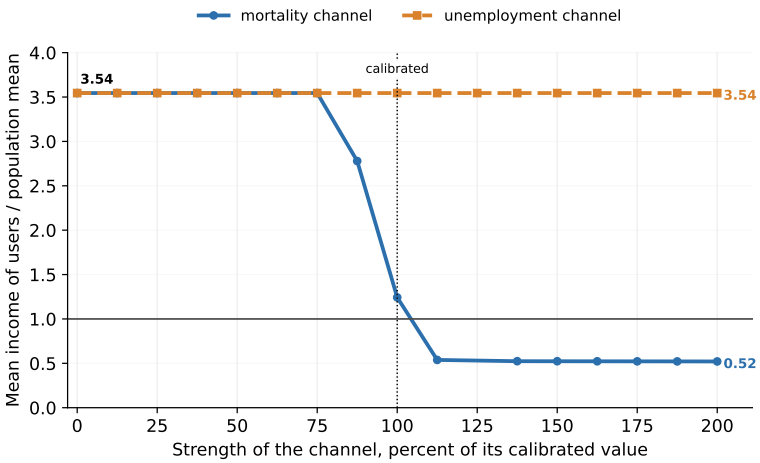



Figure 1: What it takes to reverse the income gradient in drug use

Notes: stage 1, the model without the child. Each curve scales one channel with the other switched off, so both start from the economy with no consequences of a SUD. The vertical axis is average income of users over the population average, and $b$ is recalibrated at every point so that participation is the calibrated 8.92 percent. The black dotted line marks the calibrated strength; everything to the right of it is extrapolation.

Two features of the environment are candidates to reverse it. The *mortality channel* is that the SUD increases the chance of dying, through $\kappa_{\phi d}$ in equation (4). The *unemployment channel* is that it costs the household a job with probability $u_1 - u_0$, a job loss that income support does not cover, and if its savings and food assistance cannot cover the rent then the household downsizes, consumes less, or loses their home for a shelter, which is free but whose housing services $h_1$ are below those of the small rental, $h_2$.

Figure 1 scales each channel on its own, with the other switched off, from nothing to 200 percent of its calibrated value. Scaling the mortality channel multiplies $\kappa_{\phi d}$; scaling the unemployment channel moves $u_1$ up from $u_0$, reaching its calibrated value at 100 percent. Apart from the two channels, every parameter stays at its calibrated value except $b$, which is recalibrated at every point so that participation stays at the calibrated 8.92 percent: the amount of drug use is the same everywhere on the figure, and only who uses changes. Both curves therefore start from the same economy, the one with no consequences of a SUD at all.

The result is sharper than the statement that mortality matters more. On its own, the mortality channel takes the income ratio from 3.54 with nothing switched on down to 1.24 at its calibrated strength, and falls further to 0.52 as it continues to rise to 200 percent of its calibrated value. What mortality destroys is a continuation value, and a household only stops using when the expected loss of that value exceeds what the drug is worth to it.

In contrast, the unemployment channel does not reverse the gradient at all: on its own it leaves users at 3.54 times average income at its calibrated strength, against 3.54 with nothing switched on, and at 200 percent they are still at 3.54. A resource penalty does not produce the concentration of drug use at the bottom.

Why should the two channels differ at all, when in both the disorder punishes the household? Because the cost of only one of them rises with the household's income. A job the disorder costs is not covered by income support, so it drops the household to the same floor regardless of what it was earning. What it has left is just food assistance, whatever it saved and, under Housing First, a voucher if it draws one. Because risk aversion is above one, utility has a ceiling, so how much a household loses in that fall stops growing with its income: two households well above the floor lose about the same, even if one earns twice as much as the other. The unemployment channel is therefore close to a flat charge on using, the same for every household that would use. It changes how many use, which the recalibration of $b$ undoes, but not which households do. On the other hand, buying $v$ units costs a rich household almost nothing in terms of utility and a poor household a great deal, so the utility price of using *falls* as income rises, and that, combined with

Table 3: Stage 1: the two forces and the welfare of conditionality

| | None | TF | HF |
|---|---|---|---|
| *Panel A. Outcomes, percent of households* | | | |
| Drug use | 7.62 | 4.19 | 8.92 |
| Carrying a SUD | 3.15 | 1.73 | 3.69 |
| Homeless | 0.79 | 0.56 | 0.59 |
| *Panel B. The two forces on participation, percentage points* | | | |
| Value of life, TF less none | | −3.43 | |
| Moral hazard, HF less TF | | | 4.73 |
| *Panel C. Treatment First against Housing First* | | | |
| Households strictly worse off, percent | | 10.2 | |
| Households strictly better off, percent | | 0.0 | |
| CEV of those affected, percent | | −0.17 | |
| CEV of the population, percent | | −0.017 | |

Notes: stage 1, the model without the child and without the disamenity of homelessness. None is no programme. Welfare changes are consumption-equivalent variations: the proportional change in consumption, in every period and state, worth as much as the difference in lifetime value, negative when Treatment First reduces welfare relative to Housing First.

the bounded loss from acquiring a SUD, is what puts use at the top to begin with.

Mortality is a different object. What it destroys is the whole continuation value: the consumption of a life still to be lived, and $\aleph$, the value of being alive at all, which does not shrink as consumption does. The value of life rises with permanent income, so the expected cost of using rises with income at the same time as its utility price falls. Past a point the two cross, and it is the rich who stop first. The reversal therefore happens because the mortality penalty rises quickly with income. The household with more to lose loses more, and loses disproportionally more than it gains from drugs.

## 3.5 How does housing assistance shape the drug use decision?

The two forces of question 2, moral hazard and the value of life, can be separated exactly, because they differ in what happens to a household that uses. Under Treatment First the voucher is withdrawn from any household that violates the sobriety condition, so only the value-of-life force operates. Under Housing First it is not, so both channels operate. Comparing the two policies at the same generosity therefore isolates the moral hazard force, and comparing Treatment First with no programme at all isolates the value-of-life force.

Panels A and B of Table 3 put numbers on both, in the baseline of stage 1. With no programme at all, 7.62 percent of households use. Treatment First takes that down to 4.19 percent, a value-of-life force that reduces drug use by 3.43 percentage points. Switching to Housing First then raises it even higher than before a programme existed, to 8.92 percent, so the moral hazard force raises drug use by 4.73 percentage points and the net effect of Housing First is a 1.30 percentage point increase in drug use relative to no vouchers at all. Both comparisons here are between long-run steady states, each the rest point of the economy under a policy that has always been in place. Both forces are first order and they work against each other, which is the central feature of the problem. Anyone asserting that housing the homeless must increase drug use, or must reduce it, is making a quantitative claim that needs to be tested.

The SUD rate follows participation one for one, since every user runs the same risk: 3.15 percent of households carry the disorder with no programme, 1.73 under Treatment First and 3.69 under Housing First. Homelessness falls under both programmes, from 0.79 percent to 0.56 under Treatment First and to 0.59 under Housing First.

## 3.6 Welfare

### 3.6.1 Without externalities, Housing First always wins

Sections 3.4 and 3.5 established what each policy does to behaviour. This section asks what each does to welfare, comparing the two at the same generosity $\pi_V$, in the baseline of stage 1, where nothing a household does lands on anybody else. Welfare is measured throughout as a consumption-equivalent variation (CEV), set out formally in Section 6.1. The policies differ only in the last indicator in equation (8), so Housing First transfers resources in a superset of the states in which Treatment First does. The budget set is weakly larger in every state and strictly larger in one, so a rational household with no commitment problems cannot be worse off under Housing First:

$$V_1(y;\ \mathrm{HF},\ \pi_V)\ \geq\ V_1(y;\ \mathrm{TF},\ \pi_V) \qquad \text{for every } y, \tag{12}$$

with equality exactly for households that never use under Housing First. Put the other way round, a household under Housing First can reproduce its Treatment First outcome exactly by not using: conditionality removes an option from a forward-looking optimiser and gives nothing back. This is a dominance argument, not a calibrated result, and no choice of parameters overturns it.

It follows that the moral hazard is not an inefficiency. Households here face the true SUD hazard and make no mistakes, so the extra use Housing First induces is not a wedge to be closed but a household optimally taking a risk it is able to bear. The moral hazard force of Section 3.5 is a statement about drug use and not about welfare.

Panel C of Table 3 puts welfare numbers on it: 9.4 percent of households are strictly worse off under Treatment First,[6] none is better off, and those affected lose the equivalent of 0.17 percent of consumption. Those are the households that use under Housing First and can draw a voucher: a household that never uses faces the same voucher rule under both policies and is indifferent between them.

A fixed budget does not rescue the welfare case for conditionality either. The two programmes cost almost the same, 2.14 percent of mean income per household under Treatment First against 2.15 percent under Housing First, so matching on cost rather than on generosity would change nothing. Both programmes give a voucher to the same share $\pi_V$ of eligible households, and Treatment First then denies it to those that fail the sobriety condition, so it pays out less for that reason; the two costs are close because, in equilibrium under Treatment First, households are deterred, and thus few of the households that draw a voucher fail the condition. Therefore, the case for conditionality in this model has to come from elsewhere: from what a household's drug use does to somebody else.

## 3.7 Stage 2: the child

Some of the externalities modelled in this paper are the ones imposed on the children of households with a SUD. This stage adds the child, as the quantitative model has it, and refits the model with the child in place.

**The child in the household.** Each household raises one child during its adulthood period. The child shares the household's consumption and housing through the square-root equivalence scale $\xi$, while the drug only enters the utility of the parent, so the parent and the child hold different bundles,

$$X_P = \left[\left(\frac{c}{\xi}\right)^{1-\gamma}\left(\frac{h}{\xi}\right)^{\gamma}\right]^{1-\theta}(b+v)^{\theta}, \qquad X_c = \left[\left(\frac{c}{\xi}\right)^{1-\gamma}\left(\frac{h}{\xi}\right)^{\gamma}\right]^{1-\theta} b^{\theta}, \tag{13}$$

and the parent's flow utility in period 1 becomes

$$u_1(c,h,v)\ =\ U(X_P)\ +\ \omega_c\,(\xi-1)\,U(X_c)\ +\ \aleph. \tag{14}$$

The child consumes the household's consumption and housing but none of the drug, so a parent's drug spending reaches it only by taking consumption and housing out of what they share, and it counts as $\xi - 1$

[6] The share worse off is above the 8.92 percent who use under Housing First, because a household that would use in only some of its period 1 draws is worse off in full but counts in participation only in the draws in which it uses.

of the household against the scale. The weight $\omega_c$ is the share of the child's utility that the parent takes into account. Without the drug and with $\omega_c = 1$, the first two terms are $U(X) + (\xi - 1)\,U(X) = \xi\,U(X)$, the usual form, which counts the household as $\xi$ adult equivalents: the parent as one and the child as the remaining $\xi - 1$. In period 2, when the parent moves into old age, the child leaves and the parent lives alone, $\xi = 1$, and flow utility is that of (2).

**The child leaves.** At the end of period 1 the child leaves the household and becomes an adult, entering period 1 of its own life, with its own child living with it. Its productivity is correlated with its parent's, and it is permanently reduced if the parent acquired the disorder while the child lived with it,

$$\log y_{\text{child}} \;=\; \rho \log y + (1-\rho)\log \bar{y} + \varepsilon - \delta_d\,\mathbf{1}\{d' = 1\}, \qquad \varepsilon \sim \mathcal{N}\big(0,\,(1-\rho^2)\,\sigma_y^2\big), \tag{15}$$

where $\rho$ is the persistence of productivity across generations, $\bar{y}$ is average income, the variance of $\varepsilon$ keeps the spread of productivity the same in every generation, and $\delta_d$ is the penalty for the parent's disorder.

**The parent's problem.** The child enters the parent's problem twice, once for each part of its life. In childhood it is the second term of (14). As an adult it is the continuation value $D(y)$, what the parent's disorder costs its child in the child's own lifetime value,

$$D(y) \;=\; \mathbb{E}\big[\bar{V}_1(y_{\text{child}}) \,\big|\, y,\, d' = 0\big] - \mathbb{E}\big[\bar{V}_1(y_{\text{child}}) \,\big|\, y,\, d' = 1\big] \;\geq\; 0, \tag{16}$$

where $\bar{V}_1$ is the value of entering period 1, before its draws, at the child's productivity from (15). $D(y)$ captures the direct impact of the parent's SUD on its child's utility, which runs through the child's earnings. It enters as a difference and not a level, so it costs the parent nothing unless its use makes the disorder possible. The period 1 problem in (11) in Section 3.2.4 then becomes

$$V_1(y, e, \eta) = \max_{\{k,\, v \geq 0,\, a' \geq 0\}} \Big\{ u_1(c, h_k, v) + \beta\,\mathbb{E}_{d'}\Big[\phi(d')\,V_2(a', y, d')\Big] - \;\beta\,\omega_c\,\pi(v)\,D(y) \Big\}. \tag{17}$$

$\omega_c$ multiplies both child terms. A planner who counted the child fully would put a weight of one on both, so what the parent leaves out, $1 - \omega_c$ of the child's terms, is the externality on the child. $D$ is weighted by $\pi(v)$, the chance that the use turns into a disorder. $V_1$ depends on $D$ through the parent's problem and $D$ on $V_1$ through the child's, so the two are solved as a fixed point.

**Why income, and not susceptibility.** The penalty runs through income rather than through anything the child inherits about the drug. Letting the child inherit a greater susceptibility, a higher $\pi_d$, fails. A child facing a higher chance of converting use into a SUD optimally uses *less* ex ante, which is at odds with the data: children of parents with a SUD are *more* likely to use, not less.

In contrast to the above, income works on both counts, and it works through the channels explored in Section 3.4: lower permanent income is unambiguously bad to receive, and this model already puts users at the bottom, so a child born poorer is mechanically more likely to use, which is aligned with the data. It is also supported by the empirical evidence reviewed in Section 2. The income gradient does the whole job, so the model transmits a SUD across generations without the child inheriting any greater susceptibility to the disorder.

**Calibrating the child.** With the child added, $\xi$, $\rho$, $\delta_d$ and $\omega_c$ need to be calibrated as well. I set $\xi$ to $\sqrt{2}$, the square-root scale for a household of two (OECD, 2013). The persistence of productivity across generations, $\rho$, is set to 0.50, within the range of the intergenerational elasticities of earnings in Section 2. The penalty $\delta_d$ is assigned the value 0.27 in logs to give the 24 percent earnings loss Nilsson (2017) measures for children exposed in utero to a temporary Swedish policy that let grocery stores in two regions sell strong beer, which made alcohol easier to buy during pregnancy. The parent's weight on its child $\omega_c$ is fitted so that switching the child's terms off, with every other parameter held, raises drug use by 18.3 percent, consistent with adults without children in NSDUH 2024 using 18.3 percent more than parents. Every other stage 1 parameter is refitted with the child in place, and Table 2 reports both.

**The optimal level of drug use.** To examine what the optimal degree of drug use and SUD is, one can set the weight a parent puts on its child to one, $\omega_c = 1$, so that a parent counts its child fully, with nothing else changed and the programme left as it is. In this case, no parent uses at all: participation and the SUD rate both fall to 0.00 percent, and homelessness to 0.52 percent.

**Treatment First with the child counted.** Counting the child's utility in welfare, as Section 6.1 describes, a newborn is better off under Treatment First than under Housing First, by 0.46 percent of lifetime consumption in the long run. The dominance of Section 3.6.1 still holds for the parent's own choices; what reverses the ranking is what those choices do to the child. Treatment First cuts participation from 8.93 to 7.08 percent and the share carrying the disorder from 4.00 to 2.96 percent, so fewer children enter adult life with the penalty, and the gain is 0.22 percent in childhood, because parents spend more of their resources on them instead of on drugs, and 0.24 percent in adult life, where the child enters with a better draw. Treatment First lowers homelessness too, by about 1 percent, because fewer households have a SUD and lose their jobs with no income support, and fewer are themselves poor for having been born to a parent with the disorder, so their income is higher.

## 3.8 The transition

Comparing rest points hides a key tradeoff. Normally, the debate sees Treatment First as taking immediately and giving back only later. In this model, this happens too: at implementation a household with a SUD can do nothing about its condition but loses its voucher, and it is then at a higher risk of homelessness, whereas lower substance use pays off only for the generations after it. However, in this model, a reduction in drug use can also benefit children at once: the children of parents who use drugs but do not yet have a SUD escape the earnings penalty and enjoy larger consumption while still living with their parents if their parents respond to the policy change by stopping.

The reform moves the programme from Housing First to Treatment First at the same $\pi_V = 0.27$, in the model with the child, and every generation alive when it lands is counted, each over its own remaining life and against what the old rule would have given it. Figure 2 shows the path: drug use, the disorder and homelessness of each generation as adults, from the adults at the reform on, against the Housing First rest point, and the welfare of every generation: the old and the adults alive at the reform, their children, and the cohorts born after them.

The old, in their second period, chose under the old rule, carry whatever disorder that choice left them and cannot recover in a terminal period; conditionality now withdraws the voucher from those carrying one, and they lose 0.07 percent of consumption, with 0.72 percent of them homeless against 0.71 under the old rule. The adults, in their first period with their children at home, who were born into the old distribution, choose under the new rule; they lose 0.017 percent over their adult lives, the price of an option removed.

Their children, the first cohort born into the reform, gain 0.36 percent: 0.24 in adult life, since they are the first to enter it with the draws of parents who chose under Treatment First, and 0.11 in childhood, against the 0.22 that childhood contributes in the long run, since the parents they grow up with were themselves born into the penalised distribution. From the second cohort on the chain is at its new rest point and every cohort gains the long-run 0.46 percent. The cost falls on the adults and the old alive at the reform, and the benefit on the children born at the reform and, even more, on every cohort born after them.

## 3.9 Stage 3: the disamenity of homelessness

The second externality lands on everybody. Each household's flow utility falls by $\kappa_x H$, where $H$ is the share of households that is homeless, which every household takes as given, so the term moves no choice and the calibration of stage 2 stands unchanged. It enters welfare only, and the child bears it like everyone else, once for each period.

$\kappa_x$ is solved inside the model from the willingness to pay for ending homelessness in the contingent valuation of Loubière et al. (2020), 0.2 percent of income. Households in the model spend 73 percent of their income on consumption, so 0.2 percent of income translates to 0.27 percent of consumption. $\kappa_x$ is the disamenity per unit of homelessness, in utils per period alive, that costs an adult in the stage 2 economy as

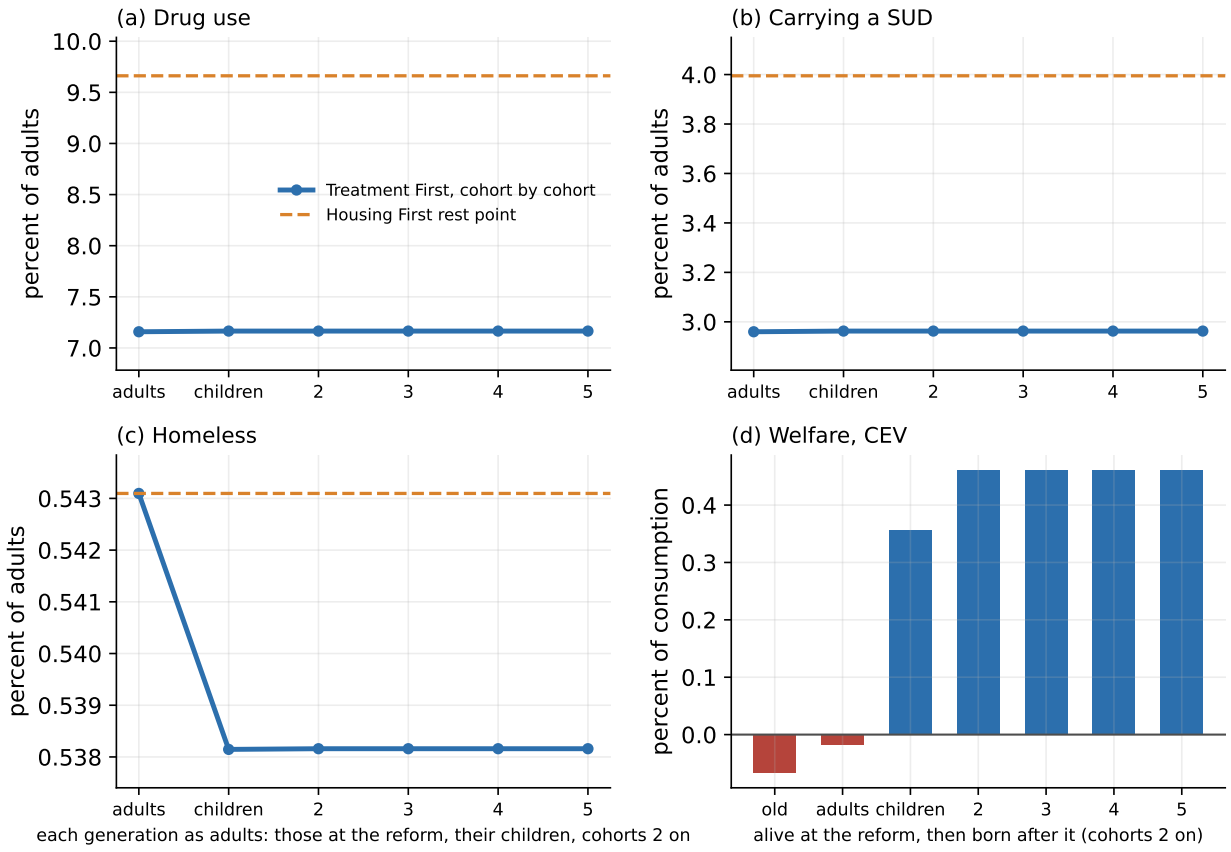



Figure 2: The transition from Housing First to Treatment First

Notes: the model with the child. Panels (a) to (c) are each generation as adults, in their first period: the adults at the reform, who were born into the Housing First distribution and choose under Treatment First, their children, and the cohorts born after them, when they become adults, to the fifth; the solid blue line is Treatment First, cohort by cohort, and the dashed orange line is the Housing First rest point. Panel (d) is each generation's CEV against Housing First: the old and the adults alive at the reform over the rest of their lives, and their children and every cohort born after them over their whole lives.

much over its adult life as giving up 0.27 percent of its consumption in every period.[7] That gives $\kappa_x = 1.36$. Because the two policies differ little in homelessness, it changes essentially nothing about the welfare gain from Treatment First, which remains at 0.46 percent of lifetime consumption.

# 4 The quantitative model

By now, I have shown how the main mechanisms in this paper work. This section describes the model the paper's numbers come from. It is very similar to the stylized model of Section 3, but includes annual periods and pieces that model leaves out: income risk over working life, retirement, recovery and treatment, SMI, a higher risk of dying while homeless, a drug test that misses some users, a government budget and a drug market. Notation follows Section 3 wherever the two models share an object: $\gamma$ and $\theta$ are the housing and drug weights in both, $\sigma$ the same risk aversion, $b$ is the same Stone-Geary shifter, $\aleph$ the same flow value of being alive, $\xi$ the same equivalence scale, $\omega_c$ the same weight on the child, $\rho$ and $\delta_d$ the same persistence of productivity and penalty, $\psi_{ui}$ and $\bar{y}$ the same replacement rate and earnings cap, $\kappa_x$ the same disamenity, $\pi_d$ the same chance that use starts a SUD, $\kappa_{\phi d}$ the same mortality multiplier for the disorder, $h_1$ the same

[7] $\kappa_x$ solves $V\big((1-w)\,c;\,0\big) = V\big(c;\,H\big)$, where $V(c;H)$ is an adult's lifetime value in the stage 2 economy when it consumes $c$ and homelessness is $H$, which it bears as the disamenity $\kappa_x H$, and $w$ is the willingness to pay as a share of consumption to have $H = 0$.

shelter, $\kappa_V$ and $\pi_V$ the same means test and chance of drawing a voucher, $p_r$ the same rent, and $\beta$ is the discount factor. The values are in Section 5.

## 4.1 Demographics and the state and choice sets

Households enter the economy as adults at model age $j = 1$, live at most $J$ annual periods, work while $j < J_R$ and are retired from $J_R$. Each has a child, who lives in the household for the first $J_K$ years of its adult life and then leaves to become an adult with a child of its own. Each period a cohort of unit mass enters at $j = 1$, made up of the grown children of the households that died, in proportion to their parents' deaths, so the mass of households is constant in a stationary equilibrium. The individual state is

$$s = (a,\, y,\, e,\, d,\, m,\, \eta) \in \mathbb{R}_+ \times \mathcal{Y}_j \times \{1,2,3\} \times \{0,1,2\} \times \{0,1\} \times \{0,1,2\}, \tag{18}$$

where $a$ denotes assets, $y$ productivity, $e$ employment status, $d$ the SUD, $m$ SMI and $\eta$ the household's standing with the voucher programme. The SUD takes three values because the parent's valuation of its child's prospects depends on its history and not only on its present (Section 4.11): $d = 0$ denotes a household that has never carried the disorder, $d = 1$ refers to one that carried it and has recovered, and $d = 2$ stands for one that carries it now. The voucher state is $\eta = 0$, denoting having no voucher, $\eta = 1$, representing holding one, and $\eta = 2$, standing for exclusion from the programme after failing Treatment First's condition.

Each period the household chooses

$$\big(h,\, v,\, \tau,\, a'\big) \in \{h_1, h_2, h_3, h_4\} \times \mathbb{R}_+ \times \{0,1\} \times \mathbb{R}_+, \tag{19}$$

where $h$ is the dwelling, of which $h_1$ is a shelter, $v$ is drug use, $\tau$ is whether or not a household is in SUD treatment, available only to a household that carries the disorder, and $a'$ is its next period's assets. Labour is inelastically supplied: a household that has a job works it.

## 4.2 Preferences

Flow utility is the stylized model's, with the child present for the first $J_K$ years. Consumption $c$ and housing $h$ are shared through the square-root equivalence scale $\xi_j$, which scales for two members while the child is at home and for one after, and the drug enters the parent's utility alone, so the parent and the child hold the bundles

$$X_P = \Big[\Big(\frac{c}{\xi_j}\Big)^{1-\gamma}\Big(\frac{h}{\xi_j}\Big)^{\gamma}\Big]^{1-\theta}(b+v)^{\theta}, \qquad X_c = \Big[\Big(\frac{c}{\xi_j}\Big)^{1-\gamma}\Big(\frac{h}{\xi_j}\Big)^{\gamma}\Big]^{1-\theta} b^{\theta}, \tag{20}$$

and period utility is

$$u\big(c,h,v,\tau\big) \;=\; U(X_P) \;+\; \omega_c\,(\xi_j - 1)\,U(X_c) \;+\; \aleph \;-\; \tau\,\bar{\tau}^{u}, \qquad U(X) = \frac{X^{1-\sigma}}{1-\sigma}. \tag{21}$$

$\gamma$ is housing's share of the non-drug bundle and $\theta$ the drug's share of the whole, $\sigma$ is the coefficient of relative risk aversion, $\aleph$ is the flow value of being alive, counted once per household, and $\bar{\tau}^{u}$ is the utility cost of a year in treatment, which is described in Section 4.6. The child's term is its own flow utility, weighted by $\omega_c$ and counted as $\xi_j - 1$ of the household; it gets none of the drug, so a parent's drug spending reaches it only through the consumption and housing they share. After $J_K$ periods the child leaves to form its own household, $\xi_j = 1$, and the child term is gone.

Aggregate homelessness $H$ carries the disamenity $\kappa_x H$ for every household and child alive, which is what makes homelessness an externality for everyone rather than only a private misfortune. However, it does not affect any household's decision: each takes $H$ as given, and the term is left out of the problem it solves. Section 5.1 takes up its size.

## 4.3 Income, taxes and transfers

**Productivity.** Log productivity follows a persistent process over working life, $z_{j+1} = \rho_z z_j + \zeta_{j+1}$ with $\zeta \sim \mathcal{N}(0, \sigma_\zeta^2)$, and it is frozen at its last working value in retirement. Earnings on the job are $y\,\varepsilon(j)$, with

$y = e^z$ normalised so that $y$ averages one at entry and $\varepsilon(j)$ the deterministic age profile. A household's productivity at entry is its inheritance from its parent, set out in Section 4.14; its spread combines $\sigma_\zeta$ with a persistent component drawn at entry, and a child entering with the penalty of that section is on a separate block of the grid, with the base grid scaled down for life.

**Before-tax income.** Before-tax income depends on age and employment,

$$Y(j,y,e) = \begin{cases} y\,\varepsilon(j) & j < J_R,\ e = 1 \quad \text{(employed)} \\ \psi_{ui}\,\min\{y\,\varepsilon(j),\ \bar{y}\} & j < J_R,\ e = 2 \quad \text{(insured, first year out of work)} \\ 0 & j < J_R,\ e = 3 \quad \text{(out of work, uninsured)} \\ S(y) & j \geq J_R \quad \text{(retired).} \end{cases} \tag{22}$$

Unemployment insurance (UI) replaces the share $\psi_{ui}$ of earnings on a base capped at average earnings $\bar{y}$, and it is paid only to a share $p_{ui}$ of those losing their job, mimicking the share of insured separations. After the first year, unemployed households, insured and uninsured alike, receive zero UI ($e = 3$). Social security $S(y)$ follows the statutory formula: a primary insurance amount that replaces declining shares of the household's productivity at its last working age between two bend points, up to the taxable maximum, the ceiling above which earnings no longer raise the benefit, and capped below by the Supplemental Security Income (SSI) payment, all in units in which $y$ averages one at entry.[8] The SUD and SMI do not cut earnings on the job; they act on whether there is a job at all, through the employment transitions below. Out of work, a household with the SUD or SMI is paid what any other household is: UI in its first year with probability $p_{ui}$, nothing with probability $1 - p_{ui}$, and nothing after the first year. There is no disability insurance in the model, so SMI brings no income of its own.[9]

**Taxes.** Before-tax income is taxed with the Heathcote et al. (2017) function and capital income at a flat rate, so that after-tax income is

$$y^n = \min\left\{Y,\ \tau_\ell\, Y^{\varphi}\right\} \ +\ (1 - \tau_k)\, r\, a, \tag{23}$$

where $\tau_\ell$ scales after-tax income, and so sets the level of the tax, and $1 - \varphi$ sets its progressivity; $\tau_k$ is the flat tax rate on capital income and $r$ is the exogenous interest rate on savings.

**Transfers.** Food assistance pays the maximum allotment $\underline{c}$ less the share $\varrho$ of income, and the voucher pays a holder that is eligible this period the rent on the dwelling above a tenant contribution of $\kappa_r$ of income, up to the fair market rent ($FMR$),

$$\begin{aligned} F &= \max\left\{0,\ \underline{c} - \varrho\,\hat{y}\right\}, \\ Vch &= \mathbf{1}\{\eta = 1\}\,\mathbf{1}\{\hat{y} < \kappa_V y^{\text{med}}\}\,\mathbf{1}\{h \neq h_1\}\,\min\left\{\max\{0,\ p_r h - \kappa_r \hat{y}\},\ \max\{0,\ FMR - \kappa_r \hat{y}\}\right\}, \end{aligned} \tag{24}$$

where $\hat{y} = Y + ra$ is the income both programmes are means-tested on, $\kappa_V$ and median income $y^{\text{med}}$ are the eligibility rule in Section 4.9, and $FMR$ is the average rent paid by housed households in the benchmark. A voucher holder in a shelter keeps the voucher and is paid nothing until it rents again, as long as it keeps its eligibility. Voucher holders keep their vouchers for as long as they remain eligible. Each entering household receives a one-time endowment $T_0$ from bequests left by deceased households, and the government collects the rest of the estates of the dead.

[8] With $\tilde{y} = \min\{y, y^{\max}\}$ the household's productivity at its last working age up to the taxable maximum $y^{\max}$, $S(y) = \max\left\{\underline{S},\ \alpha_1 \min\{\tilde{y}, k_1\} + \alpha_2 \max\{0, \min\{\tilde{y}, k_2\} - k_1\} + \alpha_3 \max\{0, \tilde{y} - k_2\}\right\}$, where $k_1 < k_2$ are the two bend points, $\alpha_1 > \alpha_2 > \alpha_3$ the shares replaced below, between and above them, and $\underline{S}$ the SSI payment. Their values are in Section 5.

[9] In the data only a minority of adults with SMI receive disability benefits on the basis of a mental disorder: about 3 million, through Social Security Disability Insurance or SSI, against 14.6 million adults with SMI in NSDUH 2023, roughly one in five.

## 4.4 The budget constraint

The household's budget constraint is as follows:

$$c + p_v v + a' \;=\; y^n + a + T_0 \mathbf{1}\{j = 1\} \;-\; p_r\, h\, \mathbf{1}\{h \neq h_1\} \;+\; Vch + F, \qquad c \geq F,\; v \geq 0,\; a' \geq 0. \tag{25}$$

Food assistance is paid in kind, so it cannot directly fund the drug, the rent or saving. The drug price is $p_v$, which is endogenous here as opposed to exogenous like in the stylized model, and the rent per unit of housing $p_r$, held at its benchmark value in every simulation. Treatment costs utility only.

## 4.5 Housing and the homelessness loop

Housing is the four point grid of (19): $h_1$ is literal homelessness, sleeping rough or in an emergency shelter, which is free, and $h_2 < h_3 < h_4$ are three rental sizes.

Homelessness then enters three separate equations: survival (29) and the employment transitions (30), both in Section 4.8, and the SUD recovery hazard (27) in Section 4.6. It raises mortality, worsens both employment margins and lowers the chance of recovering from the SUD. SMI is the one thing it does not reach, for the reason given in Section 4.7.

## 4.6 Drug use, the SUD, recovery and treatment

A household that does not carry the disorder, either $d = 0$ (never had it) or $d = 1$ (had it but recovered), and uses any quantity acquires it ($d = 2$) with probability $\pi_d$,

$$\Pr(d' = 2 \mid d \neq 2,\, v) = \pi_d\, \mathbf{1}\{v > 0\}, \tag{26}$$

and a household that has recovered can acquire the disorder again. A household may begin using only up to an age $J_E$.[10] A household with the SUD recovers, to $d = 1$, with probability

$$\begin{aligned} \pi_r(h, m, \tau) = \min \Big\{ 1,\; \bar{\pi}_r \left( \mathbf{1}\{h = h_1\}\, \mu_H + \mathbf{1}\{h \neq h_1\} \right) \left( \mathbf{1}\{m = 1\}\, \mu_m + \mathbf{1}\{m = 0\} \right) \\ \times \left( \mathbf{1}\{\tau = 1\}\, g + \mathbf{1}\{\tau = 0\} \right) \Big\}, \end{aligned} \tag{27}$$

with $\bar{\pi}_r$ the baseline annual chance of recovering while housed, without SMI and untreated, with three multipliers on it: $\mu_H < 1$, which means that recovery is harder in a shelter, $\mu_m < 1$, which means that it is harder with SMI, and $g > 1$, which means that it is more likely in treatment.

$\mu_H$ is how housing acts on drug use here. Housing does not change what a household wants: flow utility is the same function of $(c, h, v)$ housed or not. What housing changes is the chance of getting out of the disorder once it is carried. Milby et al. (2005) randomised cocaine-dependent homeless adults into arms differing in housing and measured weeks of abstinence, so the ratio between arms identifies $\mu_H$ in the calibration.

Treatment $\tau$ is a choice, taken at the utility cost $\bar{\tau}^u$ in (21), and $g$ is the only thing it buys. It does not stop the household using, it does not remove the excess mortality of the disorder while it lasts, and it does not exempt the household from Treatment First's condition, since a household in treatment still carries the disorder.

## 4.7 Serious mental illness

$m$ is a two state Markov chain representing SMI, whose prevalence is a small share of adults, defined by substantial functional impairment, rather than any mental illness, which is roughly a quarter of adults and

[10] Without it there is little consequence at old ages to using: too little life remains for the disorder to damage, so the model produces too much of it. The cause is that death is certain at the last age $J$, so close to it a household knows how little life remains, whereas in real life the vast majority of people do not know exactly when they will die. Section 5 gives the value of $J_E$.

includes mild anxiety. Onset follows an age profile $\varsigma_m(j)$, normalised to one at age 26 and scaled by the level $\omega_m$, and a household that carries the SUD is $\iota > 1$ times as likely to develop SMI as one that does not,

$$\Pr(m' = 1 \mid m = 0, d) = \min\left\{1,\ \omega_m\, \varsigma_m(j)\left(\mathbf{1}\{d = 2\}\,\iota + \mathbf{1}\{d \neq 2\}\right)\right\}, \qquad \Pr(m' = 0 \mid m = 1) = \lambda_m. \quad (28)$$

The effect of the SUD on the onset of SMI, $\iota > 1$, runs in the direction the evidence supports. About a quarter of those with a psychosis induced by a substance go on to develop schizophrenia or bipolar disorder (Murrie et al., 2020), and in the Danish registers people with any SUD are 1.34 times more likely to develop schizophrenia than people with no SUD, and those with opioid use disorder 2.74 times more likely (Nielsen et al., 2017).

SMI then affects three objects: the chances of losing a job and of finding one, through (30), mortality, through (29), both in Section 4.8, and the SUD recovery hazard, through $\mu_m$ in (27) in Section 4.6. It does not change the utility function or earnings on the job.

The SMI chain does *not* depend on housing, and that is a finding in the data rather than an omission. In the At Home / Chez Soi randomised trial, which followed 1,198 homeless adults with SMI across four cities, assignment to housing with intensive case management had no detectable effect on symptom severity at any wave.

## 4.8 What the SUD, SMI, and homelessness cost

**Death risk.** Write $1 - \bar{\phi}_j$ for the probability of dying at age $j$ of a household that is housed and has neither the disorder nor SMI; the three multipliers below raise it for every other household. Write $q_j^0(h) = \min\{1, \left(\mathbf{1}\{h = h_1\}\,\kappa_{\phi H} + \mathbf{1}\{h \neq h_1\}\right)(1 - \bar{\phi}_j)\}$ and $q_j^1(h) = \min\{1, \kappa_{\phi d}\left(\mathbf{1}\{h = h_1\}\,\kappa_{\phi H} + \mathbf{1}\{h \neq h_1\}\right)(1 - \bar{\phi}_j)\}$ for the probability without and with the disorder. The probability of dying is then

$$1 - \phi(j, h, d, m) = \min\left\{1,\ \left(\mathbf{1}\{m = 1\}\,\kappa_{\phi m} + \mathbf{1}\{m = 0\}\right) q_j^0(h) + \mathbf{1}\{d = 2\}\left[q_j^1(h) - q_j^0(h)\right]\right\}, \quad (29)$$

so $\kappa_{\phi H}$ multiplies the death probability of a household in a shelter, $\kappa_{\phi d}$ that of a household carrying the disorder, and $\kappa_{\phi m} > 1$ the baseline risk of a household with SMI, to which the excess risk of the disorder is added. A household without the disorder therefore dies with probability $q_j^0(h)$, or $\min\{1, \kappa_{\phi m}\, q_j^0(h)\}$ with SMI, and the disorder adds the excess $q_j^1(h) - q_j^0(h)$ on top. The disorder kills whether or not it is being treated.[11]

**Unemployment risk.** A job is lost with probability $\pi^{eu}$ and found with probability $\pi^{ue}$,

$$\begin{aligned} \pi^{eu} &= \min\left\{1,\ \bar{\pi}^{eu}\, \varpi(y)\left(1 + \mathbf{1}\{h = h_1\}\,\kappa_H\right)\right\}, \\ \pi^{ue} &= \min\left\{1,\ \frac{\bar{\pi}^{ue}\left(1 - \mathbf{1}\{h = h_1\}\,\kappa_H\right)}{\left(\mathbf{1}\{d' = 2\}\,\kappa_d^{ue} + \mathbf{1}\{d' \neq 2\}\right)\left(\mathbf{1}\{m' = 1\}\,\kappa_m^{ue} + \mathbf{1}\{m' = 0\}\right)}\right\}, \end{aligned} \quad (30)$$

with $\bar{\pi}^{eu}$ and $\bar{\pi}^{ue}$ the baseline separation and job finding rates, $\varpi(y)$ a factor that makes separations likelier at low productivity, in line with the unemployment rates of the education groups the productivity grid spans, and the divisors $\kappa_d^{ue}$ and $\kappa_m^{ue}$ making one harder to find with the SUD and with SMI. Neither the SUD nor SMI raises separations; each only reduces the chance of regaining employment. $\kappa_H$ does both for a household in a shelter.

**The child's productivity penalty.** A child whose parent had the disorder at any point while the child lived at home, before $J_K$ or before the parent died if that came first, enters the economy with its productivity lowered by the proportion $\delta_d$ in (39) in Section 4.14, and the penalty stays with the child for life. There is no penalty for a parent's homelessness. The parent weighs the penalty through the weight $\omega_c$ it puts on its

[11] In the data, the meta-analyses of mortality among people using extramedical opioids report overdose as the largest single contributor to the excess (Larney et al., 2020; Degenhardt et al., 2011), and in the Centers for Disease Control and Prevention (CDC) WONDER database drug poisoning alone accounts for about a fifth of all deaths at ages 25 to 44, which is the moment Section 5.3 reports the model against.

child (Section 4.11): using does not pass the penalty on by itself, but it carries the chance $\pi_d$ of acquiring the disorder, and with it the penalty. The penalty is a risk to the parent's own objective only to the extent that $\omega_c$ is positive, so it is the one risk here that carries an externality rather than a pure personal cost, which is what makes it a candidate for policy in a way that mortality and unemployment risk, both fully internalised, do not.

## 4.9 The voucher and the two policies

A household is eligible for a voucher while its income is below the share $\kappa_V$ of median income,

$$\hat{y} \;<\; \kappa_V \, y^{\text{med}}, \tag{31}$$

with $\hat{y}$ the income of (24). A holder keeps its voucher while it is eligible, so unlike Section 3.2.3 the voucher is persistent rather than redrawn each period, and an eligible household without one draws one with probability $\pi_V$. Under Housing First that is the whole rule.

Under Treatment First the programme is conditional. A household fails the condition if it uses or carries the disorder, $v > 0$ or $d = 2$, whether or not it is in treatment. Income-eligible holders and applicants that draw a voucher are tested, at a cost $c_{\text{test}}$ each to the government, and one that fails is detected with probability $p_{\text{det}}$, so unlike in Section 3.5 enforcement need not be perfect: with $p_{\text{det}} < 1$ some who fail keep their voucher. A detected holder keeps this period's payment and loses the voucher from the next period; a detected applicant never receives it. Either way the household is excluded, $\eta' = 2$: it cannot draw a voucher whatever its income, and it returns to the lottery, $\eta' = 0$, with probability $\psi_x$ each year.

None of $p_{\text{det}}$, $\psi_x$ and $c_{\text{test}}$ binds in the benchmark, where no voucher is conditional, so all three are set with the experiment in Section 6 rather than in the calibration.

## 4.10 The shocks, collected

Conditional on the state and the choices, next period's state is drawn in five transitions, which are independent given the state except that employment depends on next period's SUD and SMI and on the dwelling $h$ through (30) of Section 4.8:

**Productivity $y$.** By the process of Section 4.3 while working, and unchanged in retirement.

**Employment $e$.** A household that loses its job is insured with probability $p_{ui}$, for one year. With $\pi^{ue}$ and $\pi^{eu}$ from (30),

$$e = 1 \;\longrightarrow\; \begin{cases} e' = 1 & \text{w.p. } 1 - \pi^{eu} \\ e' = 2 & \text{w.p. } p_{ui}\,\pi^{eu} \\ e' = 3 & \text{w.p. } (1 - p_{ui})\,\pi^{eu} \end{cases} \qquad e \in \{2,3\} \;\longrightarrow\; \begin{cases} e' = 1 & \text{w.p. } \pi^{ue} \\ e' = 3 & \text{w.p. } 1 - \pi^{ue}. \end{cases} \tag{32}$$

State 2 is reachable only from employment and never from itself, so the insured are in it for the first year out of work and then move to state 3, which pays nothing.

**The SUD $d$.** A household without the disorder acquires it with the probability (26) and one with it recovers with the probability (27), both in Section 4.6, conditional on the chosen $v$, $h$ and $\tau$ and on $m$.

**Mental illness $m$.** SMI begins and ends according to (28) in Section 4.7, conditional on $d$ and age.

**The voucher $\eta$.** A voucher is drawn and revoked according to the rule of Section 4.9, conditional on this period's eligibility and, under Treatment First, on the chosen $v$ and the state $d$.

Write $\Gamma\big(s' \mid s, h, v, \tau, a'\big)$ for their joint law of motion, which is the object the expectation in (33) in Section 4.11 is taken against.

## 4.11 The household's dynamic problem

Putting it together, for $j = 1, \ldots, J$ and with $V(J{+}1, \cdot) = 0$,

$$
\begin{aligned}
V(j,s) = \max_{\{h,\,v,\,\tau,\,a'\}} \Bigg\{ & U(X_P) + \omega_c(\xi_j - 1)\, U(X_c) + \aleph - \tau\bar{\tau}^u \\
& + \ \beta\, \phi(j,h,d,m) \int_{\mathcal{S}} V(j{+}1, s')\, \Gamma\big(\mathrm{d}s' \mid s,h,v,\tau,a'\big) \\
& - \ \beta\, \chi_j\, \omega_c\, \mathbf{1}\{d \geq 1\}\, D(y) \Bigg\}, \qquad \chi_j = \begin{cases} 1 - \phi(j,h,d,m) & j < J_K \\ 1 & j = J_K \\ 0 & j > J_K, \end{cases}
\end{aligned}
\tag{33}
$$

subject to the budget (25), the income and transfer rules (22) to (24) in Section 4.3, and the restrictions that treatment is available only to a household with $d = 2$, that no household begins using after $J_E$, and that $c \geq F$, since the food stamp payment is paid in kind and cannot be saved.

The first line is the flow of (21).[12] The second is the household's own continuation. The third is the child as an adult. It enters when the child leaves: at $J_K$ with certainty, and before then with the probability that the parent dies first. $D(y)$ is what the parent's disorder costs the child, in the child's own lifetime value,

$$
D(y) = \mathbb{E}\big[V(1, s_{\text{child}}) \,\big|\, y,\ d = 0\big] - \mathbb{E}\big[V(1, s_{\text{child}}) \,\big|\, y,\ d \geq 1\big] \ \geq\ 0, \tag{34}
$$

where the child enters at $j = 1$ with the endowment $T_0$, employed, without the disorder, SMI or a voucher, and with productivity drawn as in (39) in Section 4.14, from a parent with productivity $y$ that has never had the disorder, $d = 0$, or has had it, $d \geq 1$. It is charged whenever the parent has carried the disorder at some point before the child becomes an adult.

The term is a difference and not a level, and the distinction is not cosmetic. Written as a level, $\omega_c$ times the child's value, it would be worth more to a household the likelier it is to die, because it is the death branch that carries it, and a SUD raises the death risk several fold, so a level would pay most to exactly the households the paper is about, for reasons that have nothing to do with the child. Instead, the term is zero for a parent that has never carried the disorder and a cost otherwise, so what enters the decision is only the loss the parent's own history imposes on its child.

One weight $\omega_c$ multiplies both child terms, the flow of childhood and the value of adulthood, so calibrating it pins the parent's concern for the whole of the child's life at once. At $\omega_c = 0$ the damage a household does to its child is entirely external; at $\omega_c > 0$ it internalises the share $\omega_c$ and only $1 - \omega_c$ remains external.

## 4.12 The government

The government pays social security, unemployment insurance, food assistance, vouchers and the drug tests under Treatment First. It collects the labour and capital income taxes of (23) in Section 4.3 and the estates of the dead, less the endowment $T_0$ given to entering adults ($j = 1$),

$$
\underbrace{\textstyle\sum \big(Y - \min\{Y, \tau_\ell Y^{\varphi}\}\big) + \tau_k \sum ra + \big(\text{estates} - T_0\big)}_{\text{revenue}} \ - \ \underbrace{\big(S + UI + F + Vch + \text{tests}\big)}_{\text{outlay}} \ = \ \mathcal{R}. \tag{35}
$$

In the benchmark the tax parameters are set as in Section 5 and $\mathcal{R}$ is whatever surplus they produce. In every policy experiment the taxes are held at their benchmark values and the probability of drawing a voucher $\pi_V$ adjusts so that $\mathcal{R}$ equals its benchmark value $\bar{\mathcal{R}}$; an experiment that changes a preference rather than a policy, such as $\omega_c = 1$, holds both and lets $\mathcal{R}$ move.

[12] The disamenity $\kappa_x H$ is not in the formulation, because no choice of the household changes it (Section 4.2).

## 4.13 The drug market

The price of the drug clears its market,

$$p_v = \nu_1 S_v^{1/\varepsilon_s}, \qquad S_v = \int v(s)\,\mathrm{d}\Phi(s), \tag{36}$$

where $\varepsilon_s$ is the price elasticity of supply and $\nu_1$ is normalised so that the price is one in the benchmark, and then held fixed across experiments. The price itself is not fixed: it adjusts in every experiment to clear the drug market.

## 4.14 The law of motion and newborns

Given the policy functions, the distribution over states evolves by pushing households through their choices and then through the shocks,

$$\Phi_{j+1}(\mathcal{S}') \;=\; \int \phi\big(j, h_j(s), d, m\big)\,\Gamma\big(\mathcal{S}' \mid s,\, h_j(s), v_j(s), \tau_j(s), a'_j(s)\big)\,\mathrm{d}\Phi_j(s), \tag{37}$$

for every measurable set $\mathcal{S}'$, together with the inflow of entrants

$$\Phi_1(\mathcal{S}') \;=\; \frac{\sum_{j\le J_K} \int \chi_j\,\mathcal{N}\big(\mathcal{S}' \mid s\big)\,\mathrm{d}\Phi_j(s)}{\sum_{j\le J_K} \int \chi_j\,\mathrm{d}\Phi_j(s)}, \tag{38}$$

where $\mathcal{N}$ places the child according to (39) and $\chi_j$ is the weight defined in Section 4.11. The entering cohort has unit mass every period, so in a stationary equilibrium it equals the mass of the dead. Where an entrant starts depends on its parent: its productivity is drawn around the parent's, and it carries the penalty if the parent had the disorder, both as of the moment the child becomes an adult, which is at the parent's death if that comes before $J_K$ and at $J_K$ otherwise, the same timing with which the parent weighs its child in Section 4.11.

An entrant starts with the endowment $T_0$, is employed, has never carried the disorder, holds no voucher, and carries SMI with the prevalence observed at age 26. Its log productivity is drawn around its parent's, with the persistence $\rho$ and an innovation that keeps the spread of productivity at entry the same in every generation, and it is scaled for life by the penalty if the parent had the disorder at any point before its child became an adult,

$$\begin{aligned} z_1^{\text{child}} \;&= \rho\, z^{\text{parent}} + (1-\rho)\,\mu + \epsilon \\ &\quad + \log(1-\delta_d)\,\mathbf{1}\{\text{parent has } d \ge 1 \text{ when its child becomes an adult}\}, \end{aligned} \tag{39}$$

with $z^{\text{parent}}$ the parent's log productivity when its child becomes an adult, $\mu$ the mean of log productivity at entry and $\epsilon$ normally distributed with mean zero and variance $1-\rho^2$ times the variance of log productivity at entry, which keeps that variance the same in every generation. Because the penalised child's own productivity is lower, its own child inherits a lower draw too, so the penalty reaches the grandchild through the persistence alone.

## 4.15 Competitive equilibrium

A competitive equilibrium is a collection of

- value functions $\big\{\{V_t(j,\cdot)\}_{j=1}^{J}\big\}_{t=0}^{\infty}$;
- policy functions $\big\{\{h_t(j,\cdot),\, v_t(j,\cdot),\, \tau_t(j,\cdot),\, a'_t(j,\cdot)\}_{j=1}^{J}\big\}_{t=0}^{\infty}$;
- distributions $\big\{\{\Phi_t(j,\cdot)\}_{j=1}^{J}\big\}_{t=0}^{\infty}$; and
- prices, policies and aggregates $\big\{p_{vt},\, \pi_{Vt},\, H_t\big\}_{t=0}^{\infty}$, together with $\bar{y}$, $y^{\text{med}}$ and $FMR$, which carry no date because they are kept fixed at their benchmark values,

such that, at every date $t$,

1. the policy functions solve (33) given its terminal condition, and $\{V_t(j,\cdot)\}_{j=1}^{J}$ are the associated values;
2. $\Phi_{t+1}(j{+}1,\cdot)$ follows from $\Phi_t(j,\cdot)$ by the law of motion (37) together with the entrants of (38) and (39);
3. in a policy experiment, the government budget (35) holds at $\pi_{Vt}$ with $\mathcal{R} = \bar{\mathcal{R}}$;
4. the drug market clears at $p_{vt}$, so (36) holds;
5. the homelessness rate $H_t$ is the aggregate implied by $\Phi_t$.

A stationary equilibrium is the special case in which all of these are constant and $\Phi_{t+1} = \Phi_t$, which is what the calibration and the long-run policy comparisons report. The sequence version is what the transition between them solves.

# 5 Calibrating the quantitative model

The model has fifty five parameters, some of which vary in value by age. Thirty four are calibrated externally, from data or from the literature, and twenty one are calibrated internally. Three further parameters govern drug testing and exclusion from the voucher; they do not play a role in the benchmark, where nothing is conditional, but they do play a role under Treatment First, which is where they are set. The calibrated parameters are split across the two tables of this section: twenty seven are key to the main story of this paper and its policy analysis (Table 4, fifteen parameters internally and twelve externally calibrated), and the other twenty eight are standard (Table 5, six parameters internally and twenty two externally calibrated).

The tables carry the parameter, its description, the moment and source used to fix its value, the value of that moment in the data and in the model, and the value of the parameter itself. The text alongside each table adds what a table has no room for: the intuition for what each parameter does, and how its moment identifies it. `data/DATA_APPENDIX.md` in the replication package documents every estimate in full, naming the dataset behind each target, its year and universe, and the computation that produces it. The package carries the extracts themselves alongside the code that reads them, so nothing has to be downloaded.

## 5.1 The twenty seven parameters that carry the story of this paper

Of the fifty five parameters, twenty seven are specific to this paper and carry its main story and policy analysis; the other twenty eight, common to most life-cycle models, are in Section 5.2. The former are set out in Table 4, with the parameter, its description, and the moment and source used to fix its value, which is given in the last column. They are discussed one at a time below because they are the parameters a reader may want to argue over, and each is taken in the order of the table. For clarity the table groups them into numbered blocks, each collecting the parameters that work through one mechanism, so that what each parameter does and how it moves the model can be read alongside the others that act along similar lines; the blocks are taken in turn below, and those of the standard parameters in Section 5.2 continue the numbering.

**Block 1: Who uses drugs.** The five parameters in this block set participation and, more importantly, *who* participates. Neither policy acts on them directly; they decide how much there is for a policy to act on.

**$\theta$, the weight on drugs in utility.** It is the parameter that determines the amount spent on drugs when the household consumes a strictly positive amount of it. As it rises use rises on both the extensive and intensive margins, which means that the externality rises and there is more for either policy to act on. It is calibrated internally, at 0.17, to aggregate drug spending by users over their aggregate income, 16.0 percent in the data, from retail drug spending in Midgette et al. (2019) and the incomes of adults aged 26 and over who use in NSDUH 2024.

**$b$, whether a household uses at all.** The marginal value of the drug goes as $\theta/(b{+}v)$, so a larger $b$ makes the first units worth less and a household with little to spend buys nothing. As $b$ declines use rises, raising the externality that policy can affect. Its value is chosen to target a participation rate of 8.90 percent, the NSDUH 2024 rate of a drug use disorder of any severity among adults aged 26 and over, and the model gives 8.12 percent.

Table 4: The twenty seven parameters that carry the story of this paper

| Parameter | Description | Set | Moment and source | Data | Model | Value |
|---|---|---|---|---|---|---|
| *Block 1: Who uses drugs* | | | | | | |
| $\theta$ | Weight on the drug | Int. | Drug spending over income, Midgette et al. (2019) | 16.0% | 15.3% | 0.165 |
| $b$ | Shifter on the drug | Int. | Drug participation, NSDUH 2024 | 8.90% | 8.12% | 0.0055 |
| $\pi_d$ | Chance a year of use leads to a SUD | Int. | Share of users carrying a SUD, NSDUH 2024 | 41.3% | 42.6% | 0.0345 |
| $\kappa_{\phi d}$ | Mortality multiplier, SUD | Int. | Income of drug users over average, NSDUH 2024 | 74% | 60% | 4.00 |
| $\aleph$ | Flow value of being alive | Int. | VSL, Viscusi and Aldy (2003) | 80 | 82 | 87.6 |
| *Block 2: What using and being homeless cost, and the voucher* | | | | | | |
| $\kappa_d^{ue}$ | Job-finding divisor, SUD | Int. | SUD share of the homeless, Gutwinski et al. (2021) | 21.7% | 19.2% | 3.78 |
| $h_1$ | Housing services in a shelter | Int. | Homelessness rate, HUD PiT 2023, within a band | 0.40–1.00% | 0.61% | 0.0216 |
| $\kappa_{\phi H}$ | Mortality multiplier, shelter | Ext. | Meyer et al. (2023) | 1.60 | n.a. | 1.60 |
| $\kappa_H$ | Employment penalty, shelter | Ext. | Desmond and Gershenson (2016) | 0.20 | n.a. | 0.20 |
| $\underline{c}$ | Food stamp maximum allotment | Int. | Food stamps over output (GDP), USDA | 0.42% | 0.41% | 0.0879 |
| $\pi_V$ | Chance of drawing a voucher | Int. | Recipients over eligible, HUD 2023 | 26.6% | 27.3% | 0.0675 |
| $T_0$ | Entry endowment over mean earnings | Int. | Share of homeless aged 25 to 34, HUD PiT 2024 | 26.0% | 26.6% | 0.111 |
| *Block 3: Mental illness chain* | | | | | | |
| $\iota$ | SUD multiplier on SMI onset | Ext. | Danish registers, Nielsen et al. (2017) | 2.00 | n.a. | 2.00 |
| $\lambda_m$ | Chance of recovering from SMI | Ext. | SMI ten-year recovery rate, Åsbø et al. (2022) | 50% | n.a. | 0.066 |
| $\omega_m$ | Level of the SMI onset profile | Int. | SMI share of the population, NSDUH 2024 | 4.82% | 4.82% | 0.0037 |
| $\kappa_m^{ue}$ | Job-finding divisor, SMI | Int. | SMI share of the homeless, Gutwinski et al. (2021) | 25% | 25.4% | 7.71 |
| $\mu_m$ | SUD recovery multiplier, SMI | Ext. | Dual diagnosis relapse, Bradizza et al. (2006) | 0.82 | n.a. | 0.82 |
| $\kappa_{\phi m}$ | Mortality multiplier, SMI | Ext. | Olfson et al. (2015) | 2.50 | n.a. | 2.50 |
| *Block 4: Recovery and treatment* | | | | | | |
| $\bar{\pi}_r$ | Baseline SUD recovery rate | Ext. | SUD ten-year remission rate, Vaillant (1995) | 0.025 | n.a. | 0.025 |
| $\mu_H$ | Recovery multiplier in a shelter | Int. | Abstinence, housed/homeless, Milby et al. (2005) | 1.47 | 1.47 | 0.696 |
| $g$ | Treatment multiplier on recovery | Ext. | Remission with medication for opioid use disorder | 2.00 | n.a. | 2.00 |
| $\bar{\tau}^u$ | Utility cost of treatment | Int. | Treatment take-up, NSDUH 2024 | 24.5% | 25.0% | 4.49 |
| *Block 5: Externalities* | | | | | | |
| $\omega_c$ | Weight on the child | Int. | Non-parents' use over parents', NSDUH 2024 | 18.3% | 21.8% | 0.06 |
| $\delta_d$ | Child's earnings, parent's SUD | Ext. | Earnings lost in utero, Nilsson (2017) | 0.24 | n.a. | 0.24 |
| $\rho$ | Intergenerational persistence | Ext. | Solon (1992), Mazumder (2005) | 0.50 | n.a. | 0.50 |
| $\kappa_x$ | Disamenity of homelessness | Ext. | Willingness to pay, Loubière et al. (2020) | 0.20% | 0.20% | 0.93 |
| *Block 6: Drug market* | | | | | | |
| $\varepsilon_s$ | Price elasticity of drug supply | Ext. | Mace et al. (2020) | 1.46 | n.a. | 1.46 |

Notes: the Model column is the Housing First benchmark. n.a. marks a parameter set from outside the model, which has no moment of its own to reproduce.

**$\pi_d$, the chance that a year of use leads to a SUD.** This is what makes using *risky*, and it is the step that turns drug consumption into a disorder. Raise it and the externality rises, and the moral hazard costs more, since it is $\pi_d$ that converts an extra voucher under Housing First into a higher SUD rate that Treatment First then attempts to cure. Its value is chosen so the model generates the share of users who carry a SUD, 41.3 percent among adults aged 26 and over in NSDUH 2024.

**$\kappa_{\phi d}$, increased mortality with a SUD.** As discussed at length in the stylized model, this is a key parameter in the paper. Mortality truncates the value of life for a user, and as it rises, it hits households with the highest life values hardest and thus reverses the income gradient, making the poor the users rather than the rich, in equilibrium when $b$ is chosen to match the drug use rate. It is calibrated to the average income of users, setting it to 4.00, within the range that studies of excess mortality among drug users report: from hazard ratios of 1.27 for cocaine and 2.40 for heroin among those who have ever used them (Walker et al., 2017) to a standardised mortality ratio of 14.7 among people dependent on opioids (Degenhardt et al., 2011). At that value users earn 60 percent of average income in the model, against 74 percent in NSDUH 2024.

**$\aleph$, the flow value of being alive.** $\aleph$ is the other half of the mortality mechanism: $\kappa_{\phi d}$ sets how much a SUD raises the chance of dying and $\aleph$ sets what the life being lost is worth. Increase it and use falls, because dying is more costly. Its value is chosen so the model generates a VSL of 80 times average labour income, which is the \$9m of Viscusi and Aldy (2003) against a mean household income of \$113,000.[13]

[13]No household has a negative value of life in equilibrium, so death is never an optimal outcome.

**Block 2: What using and being homeless cost, and the voucher.** Both policies work on this block, from opposite ends: each parameter says how bad a state is, so each sets what a voucher is worth to a household in that state and what being cured of that state is worth. The voucher and the endowment at entry close the block, the voucher because it is the instrument that acts on all of them. Because the child inherits a higher chance of landing in these states, everything here moves the private value and the externality together.

$\kappa_d^{ue}$**, how much harder a job is to find with a SUD.** The disorder costs earnings through whether there is a job at all rather than through what a job pays: a household carrying it finds work at the rate $\bar{\pi}^{ue}$ divided by $\kappa_d^{ue}$. Raise it and the private cost rises, the externality rises with the child's draw, and a household without a job is poorer and so likelier to be unhoused, which is the link from the disorder to the street. That is why its moment is the SUD share of the homeless, 21.7 percent in Gutwinski et al. (2021), which the model gives as 19.2.

$h_1$**, the service flow of homelessness.** It sets how bad the outside option is, and the value of being alive is measured against that outside option: a household with little to lose from dying is one for which the mortality cost of using is small. When $h_1$ rises, a shelter is less bad, so the private cost of homelessness falls, a voucher is worth less, and the externality on the child likelier to be unhoused falls with it. Its value is chosen so that the homelessness rate lies within $0.40 to 1.00$ percent of households. The floor is HUD's 2023 PiT count, which counts a single night and so misses households homeless at other times of the year; the ceiling is the share homeless at some point over a year, from Burt et al. (2001). The target is a band rather than a point because $h_1$ moves drug participation as well as homelessness, through what a household has left once it is housed. The model puts the rate at 0.61 percent. No household may choose a shelter in the last two years of life, ages 84 and 85, where, as with $J_E$ below, the end of the horizon would otherwise make the street costless.

$\kappa_{\phi H}$**, mortality when homeless.** Increase it and the private cost of being unhoused rises and thus so does the private value of a voucher. It multiplies the baseline mortality of Section 4.8 by 1.60 at every age, the mortality ratio for homeless adults in Meyer et al. (2023).

$\kappa_H$**, the employment penalty of homelessness.** This parameter decides how much more likely a household in a shelter is to lose its job and how much less likely it is to find one: separations are $1 + \kappa_H$ times as frequent and job finding $1 - \kappa_H$ times as fast. Housing First mechanically recovers earnings while Treatment First does not. As this parameter's value rises the private cost and the externality rise together, which makes it the clearest case in the table of a parameter loading Housing First's side. It is set externally at 0.20, from Desmond and Gershenson (2016).

$\underline{c}$**, the food stamp maximum allotment.** This parameter determines how low consumption can go, which equals the transfer food assistance pays at zero income. When it rises, the private cost of every bad state falls, so a voucher is worth less at the margin and the externality on a child likelier to reach those states falls too, while the budget that pays for it tightens. It is calibrated internally to food stamp spending over output, 0.42 percent: outlays from the United States Department of Agriculture (USDA) over GDP in the data, and over output, which in this model is labour income, in the model.

$\pi_V$**, the chance of drawing a voucher.** This is the policy dial itself. Increase it and the private gain rises and two externalities fall mechanically, the child's years in a shelter and the aggregate disamenity of homelessness, while the third rises through the SUD rate: the whole trade in one parameter. In the benchmark it is calibrated, at 0.068, so that 27.3 percent of the income-eligible population holds a voucher, against 26.6 percent in HUD's Picture of Subsidized Households, 2023. In every experiment it is the instrument that balances the government's budget, as set out above.

$T_0$**, the endowment at entry.** A household enters the economy with an endowment of $T_0$ times average earnings, which sets how far a young household is from the street in its first years. Homelessness at the youngest ages is steeply decreasing in what a household enters with, so this one number places the age profile of homelessness at its young end. Its moment is the share of the homeless aged 25 to 34, 26.0 percent in HUD's 2024 PiT count, which the model gives as 26.6. Raise it and the private cost of being young and poor falls, as does the child's chance of starting its adult life on the street.

**Block 3: Mental illness chain.** This block determines how much further Treatment First reaches over and above the disorder it treats: a SUD raises the chance of mental illness, mental illness lowers employment and raises death, and the child is susceptible to all of this too. Everything here therefore moves the private cost and the externality in the same direction, and the first three decide whether the chain exists at all. The chain also reproduces how much more people with SMI use drugs, which nothing in the calibration targets: in the model a household with SMI is 2.71 times as likely to use as one without, against 2.44 in NSDUH 2024 among adults 26 and over.

$\iota$**, the SUD multiplier on SMI onset.** The hinge of the whole block. Without it mental illness is exogenous to everything a policy does and treating SUD cannot touch it. As it rises a SUD causes more mental illness, so the private cost of using rises, and the externality rises further, because the child of a user now inherits a higher chance of illness that is a consequence of the productivity penalty. Treatment First's return rises as $\iota$ rises, because curing the disorder now prevents some cases of SMI too. It is set externally at 2.00, between the hazard ratios of 1.34 for any SUD and 2.74 for opioid use disorder in the Danish registers of Nielsen et al. (2017).

$\lambda_m$**, the annual chance of recovering from SMI.** This parameter governs how long an episode of mental illness lasts: it is the hazard of leaving the SMI state each year, so it sets the persistence of the illness rather than how many households ever acquire it. Raise it and mental illness is less persistent, so the private cost falls, because fewer years are spent with SMI, and the externality falls with it, since a child who is likelier to acquire a SUD and through it SMI has a higher chance of recovery and a higher value of life. It is set externally at 0.066, the annual hazard that gives a 50 percent chance of recovery within ten years, the ten-year clinical recovery rate reported by Åsbø et al. (2022).

$\omega_m$**, the level of the SMI onset profile.** This parameter governs how many households ever acquire mental illness: it scales the age profile of onset up or down without changing its shape, so it fixes the size of the SMI population rather than who reaches it or when. Raise it and both the private cost and the externality rise, since a child is likelier to reach a state that costs earnings, employment and years of life. It is calibrated internally, to the share of the population carrying SMI, 4.82 percent, which is the NSDUH 2024 prevalence by age weighted by the model's own age distribution, and the benchmark gives 4.82.

$\kappa_m^{ue}$**, how much harder a job is to find with SMI.** The same as $\kappa_d^{ue}$ but for SMI: a household with the illness finds work at the rate $\bar{\pi}^{ue}$ divided by $\kappa_m^{ue}$. Increase it and the private cost rises, because being ill costs the job, and the externality rises with the child's draw. Its moment is the SMI share of the homeless, 25 percent in Gutwinski et al. (2021), which the model gives as 25.4, for the same reason as the SUD's: a household without work is the one that loses its housing.

$\mu_m$**, the SUD recovery multiplier for SMI.** This parameter is the second direction of the chain: not only does a SUD increase the chance of SMI, but SMI also means that the SUD is less likely to be cured. As it falls the disorder is harder to recover from, so the private cost of acquiring one rises and the externality rises with it, and Treatment First has more to overcome in exactly the households it most wants to reach. It is set externally at 0.82 from the dual diagnosis relapse rates in Bradizza et al. (2006), whose review covers people carrying both a SUD and a mood, anxiety, schizophrenia-spectrum or personality disorder. Section 8 of the data appendix takes their relapse rates to the multiplier.

$\kappa_{\phi m}$**, the mortality multiplier for SMI.** The chain's last link, and the one through which it shortens life. As it rises deaths among the mentally ill rise, so the private cost rises, and the externality rises with it, because a child born to a parent with a SUD is more likely to acquire a SUD and through it SMI, and therefore more likely to die younger. This is how a SUD affects mortality indirectly, through SMI, rather than only by the direct route. It is set externally at 2.50, the middle of the standardised mortality ratios for SMI in Olfson et al. (2015).

**Block 4: Recovery and treatment.** Both policies work through parameters in this block, since recovery is what Treatment First buys directly and what Housing First buys by housing people. The first two set how much recovery there is to buy and the last two how much of it a policy can reach.

$\bar{\pi}_r$**, the baseline SUD recovery rate.** This parameter is the hazard of getting out of the disorder for a housed household, which is the level everything else in this block scales. As it rises disorders end sooner

on their own, so the private cost of acquiring one falls and the externality falls with it, and there is less for either policy to buy. It is set externally from the ten-year remission rate in Vaillant (1995), which gives an annual hazard of 0.025.

$\mu_H$**, SUD recovery when homeless.** The value of this parameter determines how much less likely recovery from the disorder is for an unhoused household than a housed one. It matters more than the level $\bar{\pi}_r$ because it is a *gradient*: it is what a voucher buys beyond shelter. As it falls (i.e., SUD recovery while unhoused is lower) the gradient steepens, so housing buys more recovery: the private value of a voucher rises and the externality falls. Near one, assistance is a transfer and the moral hazard stands unopposed. Its value is chosen so the model generates the increase in recovery housing produced in Milby et al. (2005), a randomised trial on cocaine-dependent homeless adults: an average of 6.60 weeks abstinent in the housed arm against 4.48 in the homeless, so housing raises recovery by 47.3 percent.

$g$**, the treatment effect on recovery.** This is the instrument through which Treatment First reaches households that already carry a SUD, which is what matters for short-run impacts, since on implementation some households are already in the SUD state. Raise it and treatment is more likely to cure a SUD, so a household that carries the disorder is more likely to be cured, and therefore the private cost falls, with the household choosing to go to treatment spending less of its life carrying one. The externality falls with it, because a child who acquires a disorder is also more able to address their disorder, so what its parent hands on costs it less of its own life. It is set externally at 2.00, which is how much medication for opioid use disorder raises remission: rates near double those without it, which is also what randomised trials of agonist medication against placebo find for abstinence. A behavioural adjunct such as contingency management raises the odds of recovery by far less, but it is not the treatment a policy called Treatment First would deliver.

$\bar{\tau}^u$**, the utility cost of treatment.** Take up is a choice, not a rule, and this is what makes households decline treatment they might gain from in the future. When it rises, fewer opt for treatment, so the externality rises, since more people remain with a SUD and thus more children are born with lower incomes, which leads to higher SUD incidence, and the private cost rises too, since treatment is costly. It is calibrated internally to treatment take-up: 24.5 percent of adults with a SUD received substance use treatment in the past year in NSDUH 2024, and 25.0 percent do in the model.

**Block 5: Externalities.** The parameters in this block decide the strength of the externalities. This affects the ranking between Housing First and Treatment First, since the private comparison already favours Housing First, holding the externalities aside. The first three are how much of the child a parent counts, what a parent's disorder costs the child and how much of a parent's productivity the child keeps, and the fourth is what the homelessness of strangers costs everybody. All four concern externalities in the strict sense: one household's choice enters another household's utility directly without being internalised by the household making the decision, and it would still do so even if the programmes were paid for with a lump sum tax.

$\omega_c$**, the parent's weight on its child.** A parent bears $\omega_c$ of what its choices cost its child, in the childhood they share and in the draw the child enters with, and $\omega_c$ is small. As it rises the parent counts more of the child's loss, so the externality left for a policy to correct is smaller. Identifying it requires care, because *every* household here raises a child, so there is no parent against non-parent comparison to run inside the benchmark. It is identified instead by a counterfactual moment: with every other parameter unchanged, drug use with the child given no weight is compared with drug use at $\omega_c$. In NSDUH 2024 adults aged 26 and over without children use at a rate 18.3 percent above parents', read the way the model reads it, and at $\omega_c = 0.06$ removing the child's weight raises use by 21.8 percent in the model, from 8.12 to 9.89 percent.

$\delta_d$**, what the child loses to a parent's SUD.** The child does not inherit the disorder, only a worse productivity draw, and that lower expected draw is what makes it likelier to be poor, to use, to acquire a disorder and mental illness consequently, to be unhoused and to die earlier. Increase it and the externality rises, and it loads whichever policy prevents more disorder: Treatment First by curing the parent, Housing First because recovery is easier once housed, while the parent's own private cost is largely unchanged, which is exactly what makes it an externality. It is set externally at 0.24, the

earnings loss Nilsson (2017) measures for children exposed in utero to a Swedish alcohol experiment. The loss is keyed on whether the parent carries the disorder when it dies, which is the date the child enters, and it lasts the child's whole life.

$\rho$**, the persistence of productivity across generations.** The child's log productivity at entry is drawn around its parent's with weight $\rho$ and around the mean with weight $1-\rho$, so $\rho$ is how much of its parent's standing a child keeps. It is the channel through which anything that lowers a parent's productivity lowers its child's, and so the channel through which a policy that keeps a parent in work reaches the next generation. Raise it and a child's prospects depend more on its parent's, so both externalities that run through the draw grow. It is set externally at 0.50, between the intergenerational elasticities of earnings in Solon (1992) and Mazumder (2005).

$\kappa_x$**, the disamenity of homelessness.** This is the only parameter through which one household's housing status is connected to the utility of another who is not its own child. It enters welfare and not choices: each household's flow utility falls by $\kappa_x$ times the aggregate homelessness rate, which no household can move on its own, so it moves no policy function and prices only what the policies do to the headcount. As it rises the externality from the aggregate headcount grows, which loads whichever policy houses more people. Setting it high should mostly favour Housing First, so the value used errs on the high side, and a ranking that still favours Treatment First survives an assumption unkind to it. The estimate is the willingness to pay for ending homelessness in the contingent valuation of Loubière et al. (2020), 0.20 percent of income, the upper end of its estimates. The value used, 0.93 in units of utility, was set to it, and at the benchmark it is worth 0.30 percent of consumption, the same willingness to pay measured against consumption rather than income.

**Block 6: Drug market.** One parameter of the drug market belongs here. It is what makes the drug price endogenous, so drug use affects drug demand and the drug price in equilibrium. This is a general-equilibrium offset that runs against Treatment First in particular. Curing households takes them out of the drug market, demand falls and so does the price, and a cheaper drug draws other households in. Some of what the policy removes is handed back through the price, and how much depends entirely on this parameter.

$\varepsilon_s$**, the price elasticity of drug supply.** I set it to 1.46, which is what Mace et al. (2020) estimate for cannabis on Washington State administrative data. To the best of my knowledge, there is no direct estimate for an illicit drug, and cannabis is the closest market there is: it remains a Schedule I substance federally, so even a state licensed producer works without interstate trade, without ordinary banking and under federal prohibition, which is the set of frictions an illicit supplier faces. When it rises, supply is more elastic, so the price moves less when demand does, and the general-equilibrium offset to treatment shrinks: at a very elastic supply, curing households barely lowers the price, so general-equilibrium effects play little role.

## 5.2 Standard parameters

The remaining twenty eight parameters are more conventional, and I record the values I use so that the calibration can be reproduced. Six of them I calibrate inside the model, against a moment, because no source reads them off directly; the other twenty two I set from outside, from an estimate or from statute. Table 5 and the entries below run in the same seven blocks, and each entry gives the moment or the source that fixes the parameter together with the value that moment takes in the data. The Model column reports what the model produces against that value, and it does so only for the six I calibrate internally, since the other twenty two have no target to reproduce.

These parameters are not inert in the experiments. For example, the discount factor governs how heavily a household weighs a death later in life, the dispersion of productivity determines how many households are poor enough to be at risk of homelessness and of acquiring a disorder, and risk aversion governs what insurance is worth. What separates them from the parameters above is not that they do nothing, but that they are not specific to this paper.

Table 5: Standard parameters

| Parameter | Description | Set | Moment and source | Data | Model | Value |
|---|---|---|---|---|---|---|
| *Block 7: Preferences* | | | | | | |
| $\sigma$ | Risk aversion | Ext. | Attanasio and Weber (2010) | 2.00 | n.a. | 2.00 |
| $\gamma$ | Housing weight, non-drug bundle | Int. | Housing over expenditure net of drugs, CEX 2024 | 26.8% | 26.0% | 0.536 |
| $\beta$ | Discount factor | Int. | Wealth over output, BEA fixed assets | 3.00 | 2.94 | 0.946 |
| $\xi_j$ | Equivalence scale, child at home | Ext. | Square-root scale, OECD | 1.41 | n.a. | 1.41 |
| *Block 8: Rental ladder* | | | | | | |
| $h_2$ | Smallest rental size | Int. | Renters paying over 30% of income, SCF 2022 | 43.2% | 58.9% | 0.0407 |
| $h_3/h_2$ | Middle rung over the smallest | Ext. | Set | 4.00 | n.a. | 4.00 |
| $h_4/h_2$ | Largest rung over the smallest | Ext. | Set | 8.00 | n.a. | 8.00 |
| *Block 9: Earnings and employment* | | | | | | |
| $\varepsilon(j)$ | Age profile of efficiency | Ext. | Hump peaking at age 56, Greenwood et al. (2025) | profile | n.a. | profile |
| $\rho_z$ | Persistence of productivity | Ext. | Set | 0.90 | n.a. | 0.90 |
| $\sigma_\zeta$ | Standard deviation of its shocks | Int. | Bottom half's share of earnings, SCF 2022 | 17.2% | 16.3% | 0.413 |
| $\bar{\pi}^{eu}$ | Job separation rate | Int. | Unemployment rate, BLS CPS 2014–24 | 4.9% | 5.6% | 0.0268 |
| $\bar{\pi}^{ue}$ | Job finding rate, annual | Ext. | Krueger et al. (2014) | 0.55 | n.a. | 0.55 |
| $p_{ui}$ | Share of separations insured | Ext. | Set | 0.30 | n.a. | 0.30 |
| *Block 10: Survival and mental illness* | | | | | | |
| $\bar{\phi}_j$ | Baseline survival by age | Ext. | SSA period life table, ages 26 to 85 | profile | n.a. | profile |
| $\varsigma_m(j)$ | Age profile of SMI onset | Ext. | SMI prevalence by age, NSDUH 2024 | profile | n.a. | profile |
| *Block 11: Interest and taxes* | | | | | | |
| $r$ | Risk-free rate | Ext. | Kaplan et al. (2020) | 0.03 | n.a. | 0.03 |
| $\varphi$ | Tax progressivity | Ext. | Heathcote et al. (2017) | 0.85 | n.a. | 0.85 |
| $\tau_k$ | Capital income tax | Ext. | US long-run rate on gains and dividends | 0.15 | n.a. | 0.15 |
| $\tau_\ell$ | Tax level | Int. | Average labour income tax, McDaniel (2007) | 22.4% | 21.1% | 0.894 |
| *Block 12: Social security and programmes' rules* | | | | | | |
| $S(y)$ | Replacement rates, bend points | Ext. | SSA primary insurance amount | statute | n.a. | 0.90, 0.32, 0.15 |
| $\underline{S}$ | SSI floor over entry productivity | Ext. | SSA federal benefit rate | 0.177 | n.a. | 0.177 |
| $\psi_{ui}$ | UI replacement rate | Ext. | DOL statutory replacement rates | 0.50 | n.a. | 0.50 |
| $\bar{y}$ | Cap on the UI earnings base | Ext. | DOL state benefit maxima | 1.00 | n.a. | 1.00 |
| $\kappa_V$ | Voucher means test | Ext. | HUD rule | 0.50 | n.a. | 0.50 |
| $\kappa_r$ | Tenant's rent, share of income | Ext. | HUD rule | 0.30 | n.a. | 0.30 |
| $\varrho$ | Food stamp taper rate | Ext. | SNAP statute, 7 CFR 273.10 | 0.30 | n.a. | 0.30 |
| *Block 13: Drug market* | | | | | | |
| $J_E$ | Last age use can begin | Ext. | Model age 55, that is age 80 | 55 | n.a. | 55 |
| $\nu_1$ | Drug supply shifter | Ext. | Normalisation, so $p_v = 1$ | norm. | n.a. | norm. |

Notes: the Model column is the Housing First benchmark. n.a. marks a parameter set from outside the model, which has no moment of its own to reproduce.

**Block 7: Preferences.** The four parameters in this block say what a household wants, and what it is willing to give up to get it.

**$\sigma$, risk aversion.** I set it to the standard value of two, following Attanasio and Weber (2010). Increase it and households want smoother consumption and insurance is worth more, so the voucher is worth more; it also sets the marginal utility of consumption, so it enters every welfare comparison in the paper.

**$\gamma$, the weight on housing services.** This is the share of housing in the bundle of consumption and housing that the drug is then weighed against, so it fixes what a dwelling is worth against consumption. Because housing comes in four sizes, the shelter included, rather than any size, the share of spending that goes on housing is not $\gamma$ itself, as it would be with a continuous choice, and $\gamma$ is chosen for the share it produces: 26.8 percent of expenditure, as shelter plus utilities take in the 2024 Consumer Expenditure Survey, which records no illicit drugs; the model's share is of expenditure including them. At its value of 0.54, the model's share is 26.0 percent. As $\gamma$ rises housing is worth more against consumption, households buy more of it, and that expenditure share rises with it.

**$\beta$, the discount factor.** It is chosen so that wealth over output matches the national accounts, a ratio of three, from the Bureau of Economic Analysis (BEA) fixed asset tables against National Income and Product Accounts (NIPA) output, both read at the 2023 vintage, the last year the fixed asset tables cover. As it rises households are more patient, save more, and the wealth to output ratio rises with it.

**$\xi_j$, the equivalence scale.** The household's consumption and dwelling are shared with the child while it lives at home, and they are divided by $\xi_j$ to give what each member gets. It is the square-root scale,

$\sqrt{2}$ for a household of two, from the parent's age 26 to 50, and one after the child has left.

**Block 8: Rental ladder.** These three parameters build the grid of dwelling sizes a household chooses from. Below the smallest rung is the shelter, $h_1$, in Table 4.

$h_2$**, the smallest rental size.** The size of the cheapest dwelling is what decides how much of its income a poor household must spend to be housed at all, so its moment is the share of renters paying more than thirty percent of their income for housing, 43.2 percent in the 2022 SCF. Raise it and the bottom rung is larger and dearer, so more households are burdened. It is calibrated to this moment, and the model overshoots the share, at 58.9 percent.

$h_3/h_2$ **and** $h_4/h_2$**, the spacing of the ladder.** The two larger rungs are set at 4.00 and 8.00 times the smallest. The share of the housed in the largest unit size, which a calibrated top rung would answer to, is reported with the untargeted moments instead.

**Block 9: Earnings and employment.** These six parameters determine what a household is expected to earn over its life, how persistent the shocks to it are, and how it moves between work and unemployment.

$\varepsilon(j)$**, the age profile of efficiency.** The deterministic hump in earnings over the life cycle, taken as the exponential of a quadratic in age used by Greenwood et al. (2025), $\varepsilon(j) = \exp\{-(j-1)^2/1800 + (j-1)/30\}$ with $j = 1$ at age 26. It peaks at age 56, having risen 65 percent from age 26, and then falls 4 percent from the peak to age 65, the last working age. Without it every household would earn the same at every age, and the ages at which households are poor, and so at risk of homelessness and of using, would be wrong at both ends of working life.

$\rho_z$ **and** $\sigma_\zeta$**, the productivity process.** Log productivity is persistent but not permanent: it follows an autoregressive process with persistence $\rho_z = 0.90$ and shocks of standard deviation $\sigma_\zeta$, on 13 nodes, and it is frozen at its last working value in retirement. The spread at entry combines that scale with a persistent component of the entry draw, set with a standard deviation of 0.31 before the common scale that $\sigma_\zeta$ carries, 0.64 after it, and a persistence of 0.50, which gives a standard deviation of 0.84 in logs. The grid spans 2.69 standard deviations below the mean and 3.00 above, with its lowest node at 0.009 of mean productivity, and separations are tilted across it, through $\varpi(y)$, by the unemployment rates of four education groups in the Bureau of Labor Statistics (BLS) 2019 annual averages for ages 25 and over. $\sigma_\zeta$ is what decides how many households are poor enough to be at risk of homelessness and of acquiring a SUD, and it is calibrated to the share of earnings that goes to the bottom half of working households: in the 2022 SCF, households aged 26 to 65 with positive wage and salary income in the poorer half of the distribution earn 17.2 percent of total earnings. At its value of 0.41, the model gives 16.3 percent. The share falls in $\sigma_\zeta$.

$\bar{\pi}^{eu}$**, the job separation rate.** Chosen to reproduce the aggregate unemployment rate in the BLS Current Population Survey (CPS) over 2014 to 2024, 4.9 percent, given the job finding rate set externally below; at its value of 0.027 the model gives 5.6. It is scaled by $\varpi(y)$ in (30), so separations are likelier at low productivity. Raise it and more households lose work each year, so the unemployment rate rises with it.

$\bar{\pi}^{ue}$**, the annual job finding rate.** The probability that an unemployed household finds work by the end of a period, which I set to 0.55 from the annual exit rate Krueger et al. (2014) estimate on matched CPS panels.

$p_{ui}$**, the share of separations that are insured.** A household that loses its job draws UI for one year with probability 0.30 and nothing otherwise. It sets how many of the newly unemployed fall straight to zero earnings.

**Block 10: Survival and mental illness.** These two parameters are age profiles that I take from outside the model rather than generate inside it.

$\bar{\phi}_j$**, baseline survival by age.** The most recent SSA period life table, Table 4.C6, for ages 26 to 85, with the sexes combined by survivor weight rather than averaged. This is the profile every mortality multiplier

in Section 5 multiplies, so it is the base the SUD, homelessness and mental illness hazards are measured against.

$\varsigma_m(j)$**, the age profile of SMI onset.** The shape of onset across ages, the onset that reproduces a log-linear fit to the share of adults carrying SMI at each age in NSDUH 2024, normalised to one at age 26, so that $\omega_m$ in Table 4 sets its level alone. It sits here rather than there because it is the baseline profile, in the same way the life table is: what this paper adds is $\iota$, the multiplier a SUD puts on that onset, and that is where the interaction lives.

**Block 11: Interest and taxes.**

$r$**, the risk-free rate.** I set it to three percent, from Kaplan et al. (2020).

$\varphi$**, tax progressivity.** The exponent on income in the tax schedule of Section 4.3 is set to 0.85 from Heathcote et al. (2017). It determines the progressivity of the labour income tax. That in turn decides how much a programme's bill falls on the households it is meant to help, and how much on everybody else.

$\tau_k$**, the capital income tax.** I set it to fifteen percent, consistent with the statutory United States rate on long-term capital gains and qualified dividends for the middle of the income distribution, which the Internal Revenue Service (IRS) sets out in Topic 409 (`https://www.irs.gov/taxtopics/tc409`).

$\tau_\ell$**, the level of the labour income tax.** It is calibrated so that the average labour income tax rate the model produces is 22.4 percent, which is what McDaniel (2007) measures for the United States and what Guvenen et al. (2023a) use; at its value of 0.89 the model gives 21.1 percent. It does not move in the experiments: the budget is balanced by $\pi_V$ instead.

**Block 12: Social security benefits and programmes' rules.**

$S(y)$**, social security.** Retirement income follows the statutory primary insurance amount: it replaces $0.90, 0.32, 0.15$ of the household's productivity at its last working age below the first of the SSA's two bend points, between them and above the second, up to the taxable maximum, with the Supplemental Security Income payment as a floor. The formula and the floor apply to productivity, in units in which it averages one at entry, so the floor, 0.18 in those units, is 0.11 of the benchmark's average labour income. As it rises the old are richer, so the value of a life still to be lived rises and with it the cost of a disorder that shortens it.

$\psi_{ui}$ **and** $\bar{y}$**, unemployment insurance.** These two set the UI replacement rate and the cap on the earnings it is paid on. I set them to fifty percent and one times average labour income, respectively, taking both from the statutory replacement rates and state benefit maxima that the Department of Labor compiles in its comparison of state UI laws (`https://oui.doleta.gov/unemploy/statelaws.asp`).

$\kappa_V$ **and** $\kappa_r$**, the voucher rules.** These two set the income cutoff for eligibility and the tenant's contribution, and I take both from HUD's own rules. I set them to half of median income and thirty percent, respectively. The first is the very low-income limit HUD publishes annually with its income limits (`https://www.huduser.gov/portal/datasets/il.html`); the second is the share of its income a tenant pays towards the rent, the voucher paying the rest up to the payment standard, under the Housing Choice Voucher programme (`https://www.hud.gov/helping-americans/housing-choice-vouchers`).

$\varrho$**, the food stamp clawback rate.** This is the rate at which food assistance is withdrawn as income rises. I set it to thirty percent, which is the statutory rule: a household's monthly allotment is the maximum allotment for its size reduced by thirty percent of its net monthly income (7 CFR 273.10). It matters beyond the level of the transfer, because it is the implicit marginal tax the programme puts on earnings for the households the paper is about: an extra dollar earned costs a household on food assistance $\varrho$ of a dollar of its allotment.

**Block 13: Drug market.** These two parameters describe the rest of the drug market, closing the loop on the elasticity explained above: the first fixes who can enter it, and the second fixes the units in which its price and quantity are measured.

$J_E$**, the maximum age at which use can begin.** This restricts only households that do not have a SUD, not ones that already do. I set it to fifty five in model terms, which corresponds to age 80 in the data. It is there so that use cannot begin in the last few years of life, where almost no continuation value remains to be lost and the private cost of using is therefore close to zero. That is an artefact of the horizon rather than a feature of the world: to keep the age dimension manageable the model imposes certain mortality after age 85, so a household near the end knows death is coming with certainty and a disorder costs it almost nothing. Without the cap the model would let use begin exactly there, and at the final age it would be worse than that, since a household that acquires the disorder in the last period dies at the end of it whatever the hazard does: the disorder carries literally no consequence and only the flow benefit of using, which cannot be a reasonable description of the choice either.

$\nu_1$**, the supply shifter.** A pure normalisation, chosen so that supply equals demand at $p_v = 1$ in the benchmark, and kept fixed across experiments, in which the price is free. It normalises units, not behaviour.

## 5.3 Validation

None of the moments in Table 6 is fitted, with one qualification: the share of the homeless in the youngest band is the target of $T_0$. Each is produced endogenously by a model with sixty single ages, and each is a place the model can be caught. Section 9 of `data/DATA_APPENDIX.md` in the replication package derives the data side of each, and it is the place to go for everything needed to replicate them: the dataset, the year, the universe, the computation, and the three cases where the universe rather than the arithmetic decides the number. The package ships those datasets as well, so none of them has to be downloaded first.

Table 6 is read below one block at a time.

**Homelessness.** The first row gives each age band's share of the homeless. The youngest band is the target of $T_0$; the other four are free. The model puts 38.1 percent of the homeless at ages 35 to 44, against 27.2 in HUD's PiT count, and 11.2 percent at 55 and over, against 25.9. Of the homeless in the model, 21.4 percent are employed, against 46 in the data, and their income is 0.2 percent of the average, against 19.1 in Meyer et al. (2024).

**Who uses and who carries a SUD.** Users in the model are younger than in the data: 67 percent of them are 26 to 34, against 45 in NSDUH 2024, and their mean age is 34.6, against 39.6. Unemployment among households with a SUD is 14.7 percent, against 16.2. Households with SMI are 2.71 times as likely as others to use drugs, against 2.44, and drug poisonings account for 13.3 percent of deaths at ages 25 to 44, against 20.0 in CDC WONDER.

**Treatment.** The rows from the Treatment Episode Data Set for Admissions (TEDS-A) describe admissions to publicly funded treatment, so they are read as shares within admissions rather than as levels. Of admissions in the model, 3.2 percent are homeless, against 14.5 in the data.

**Housing and vouchers.** Of renters in the model, 26.8 percent spend over half their income on rent, against 19.2 in the SCF, and 86.5 percent of the housed live in the largest unit size, against 62.7 in the American Community Survey. A voucher holder in the model pays 38.4 percent of its income in rent, against 31.2 in HUD's Picture of Subsidized Households, and 39.8 percent of voucher holders live mainly on wages, against 28.0.

**Distribution.** The poorest fifth of households receive 3.40 percent of income in the model, against 2.71 in the SCF. Their share of net worth is 0.29 percent, against $-0.23$, because in the SCF the bottom of the distribution owes more than it owns, and the bottom half holds 6.4 percent of net worth, against 2.5 in the Federal Reserve's Distributional Financial Accounts.

Table 6: Untargeted moments

| Moment | Data | Model | Source |
|---|---|---|---|
| *Homelessness* | | | |
| Share of homeless by age, 25–34 to 65+[a] | 26.0, 27.2, 21.0, 18.4, 7.5 | 26.6, 38.1, 24.1, 10.4, 0.8 | HUD PiT 2024 |
| Employed share of the homeless | 46 | 21.4 | HUD PiT 2023, TEDS-A 2022 |
| Income of the homeless, over average | 19.1 | 0.2 | Meyer et al. (2024) |
| *Who uses and who carries a SUD* | | | |
| Share of users by age, 26–34 to 65+ | 45, 19, 14, 12, 10 | 67, 16, 10, 5, 2 | NSDUH 2024 |
| Mean age of users, years | 39.6 | 34.6 | NSDUH 2024 |
| Unemployment given a SUD | 16.2 | 14.7 | NSDUH 2024 |
| Drug use with SMI over without, ratio | 2.44 | 2.71 | NSDUH 2024 |
| Drug deaths as a share of deaths, 25–44 | 20.0 | 13.3 | CDC WONDER |
| *Treatment* | | | |
| Homeless at treatment admission | 14.5 | 3.2 | TEDS-A 2022 |
| Psychiatric problem at SUD admission | 43.1 | 0.1 | TEDS-A 2022 |
| Working at all, homeless in treatment | 6.6 | 0.0 | TEDS-A 2022 |
| *Housing and vouchers* | | | |
| Renters paying over half of income | 19.2 | 26.8 | SCF 2022 |
| Housed in the largest unit size | 62.7 | 86.5 | ACS 2023, B25042 |
| Rent over income, voucher holders | 31.2 | 38.4 | HUD PSH 2023 |
| Voucher holders living on wages | 28.0 | 39.8 | HUD PSH 2023 |
| *Distribution, shares of the total* | | | |
| Income, bottom 1, 5, 10, 20% | 0.01, 0.31, 0.92, 2.71 | 0.01, 0.19, 1.01, 3.40 | SCF 2022 |
| Net worth, bottom 1, 5, 10, 20, 50% | −0.16, −0.28, −0.29, −0.23, 2.5 | 0.00, 0.00, 0.01, 0.29, 6.4 | SCF 2022, FRB DFA |
| *Transmission, odds ratios* | | | |
| Child of a SUD parent acquires a SUD | 1.44, 2.51 | 1.50 | Mellentin et al. (2016), Yoon et al. (2013) |
| Homeless adult had a homeless caretaker | 1.8 to 2.4 | 4.92 | Bassuk et al. (1997) |

Notes: percent, except the transmission block, which is ratios; the model is the Housing First benchmark. [a]The share aged 25 to 34 is the target of $T_0$ in Table 4, so of this row only the other four bands are untargeted.

**Transmission.** A child of a parent with a SUD is 1.50 times as likely to acquire one in the model, within the range of 1.44 to 2.51 in Mellentin et al. (2016) and Yoon et al. (2013). An adult who is homeless in the model is 4.92 times as likely to have had a homeless caretaker, above the 1.8 to 2.4 of Bassuk et al. (1997), and that row needs a word about its construction. Bassuk et al. (1997) is a case-control study of 220 homeless mothers and 216 low-income housed mothers, so both arms are poor and the comparison isolates transmission from the poverty that accompanies it. The share whose primary female caretaker had ever been homeless was 14.2 against 8.3 percent, and for the male caretaker 10.3 against 4.4, which, computed on the two samples' counts, are odds ratios of 1.8 and 2.4. A ratio taken against the general population instead would be far larger and would credit the housing penalty with everything else that differs between poor children, which Cobb-Clark and Zhu (2015) show is substantial: among the households of the Journeys Home panel (Scutella et al., 2012), every one of them on income support, those first homeless as children were more than twice as likely to have been in foster care.

# 6 Experiments and results

The experiment is the reform the introduction describes. The programme stays rationed and only its rule changes, from Housing First, the benchmark, to Treatment First, under which a household that uses or carries a SUD cannot hold a voucher (Section 4.9). Every parameter Section 5 calibrates keeps its benchmark value, the weight a parent puts on its child and the disamenity of homelessness included, and so do the aggregates the government's rules are indexed to. Two prices move. The probability of drawing a voucher $\pi_V$ adjusts so that net government revenue stays at its benchmark level, since conditionality changes who holds a voucher and so what the programme costs, and the drug price clears the drug market, with the labour tax held where it is.

**The screen.** Under Treatment First a household fails the condition if it uses drugs in the period or carries the disorder, and treatment does not exempt it. A holder that fails is caught with probability $p_{\text{det}}$ and loses its voucher, and an applicant that draws a voucher is screened the same way before it receives one. A household that is caught is excluded from the programme whatever its income, and returns to the queue for a voucher with probability $\psi_x$ a year. The government pays $c_{\text{test}}$ for every household it screens, inside

the budget that $\pi_V$ balances. None of these three parameters does anything in the benchmark, where no voucher is conditional, so they are set here rather than in Section 5.

The detection rate is $p_{\text{det}} = 85$ percent, chosen rather than measured, since no programme publishes a detection rate for a drug screen in a housing programme, and no household that meets the condition is flagged. The release rate is $\psi_x = 1/3$, an expected exclusion of three years. The test costs $c_{\text{test}} = 0.019$ percent of average earnings, below the assay cost in the state fiscal notes of the public assistance programmes that have tried testing, which is itself a lower bound on the cost of administering a screen once staffing is counted.

## 6.1 The welfare criterion

Every welfare number in this paper is a CEV, and the convention matters enough to state exactly. Write $V^b$ for lifetime value under the benchmark and $V^c$ for lifetime value under a counterfactual policy. The CEV $\Delta$ is the proportional change in consumption, applied at every date and in every state, that makes a newborn under the benchmark as well off as the same newborn under the counterfactual:

$$V^b\big((1+\Delta)\,c\big) \;=\; V^c(c). \tag{40}$$

A positive $\Delta$ therefore means the counterfactual is worth that much of a permanent consumption gain, and a negative one that it costs that much.

**Whose welfare.** The household considered is a newborn, and its welfare counts the whole of its life: its childhood in its parent's household and its adult life after it. The childhood part is the child's own flow utility from the bundle it shares with its parent, counted as $\xi - 1$ of the household as it is in the parent's problem and with the flow value of being alive, $(\xi-1)\big[U(X_c)+\aleph\big]$, less the disamenity of aggregate homelessness $\kappa_x H$ that it bears like everyone else, averaged at each age over the households of that age, summed over the years it lives at home and discounted to its birth; the adult part is its value on entering the economy at age 26, discounted by the same years. Its own childhood counts at weight one here, not at its parent's weight $\omega_c$, and that is what brings the externality on the child into welfare; its adult part is the value its own problem maximises, so the child it raises in turn enters there at the weight $\omega_c$. The value is taken before any of its draws are realised, including the household it is born into, so $V^b$ and $V^c$ are expectations over the productivity draw and everything that follows from it, and $\Delta$ equates them. The expectation is taken under each policy's own equilibrium rather than a common one, since the penalties that set a child's starting productivity depend on what happened to its parent, so the distribution a newborn is drawn from is itself something the policy moves. This is the standard criterion in this literature (Conesa et al., 2009; Guvenen et al., 2023b).

Two horizons are used, and they are the same object measured at different dates: a newborn on the date the policy is implemented, who lives through the transition, and a newborn in the new steady state. Because both are the ex-ante value of a child at birth, childhood included, any difference between them is the transition itself rather than a change in who is being counted.

**How $\Delta$ is computed.** In both models consumption enters every bundle, the parent's and the child's, with the exponent $(1-\gamma)(1-\theta)$, so scaling consumption by $1+\Delta$ at every date multiplies every term of the form $U(X)$ by $(1+\Delta)^{(1-\gamma)(1-\theta)(1-\sigma)}$, while the flow value of being alive $\aleph$, which is additive, does not scale at all. Writing $A$ for the discounted lifetime count of every term that does not contain consumption, which is $\aleph$ and the disamenity $\kappa_x H$ and, in the quantitative model, also the utility cost of treatment and the parent's charge for its child's inherited penalty, $V\big((1+\Delta)c\big) = (1+\Delta)^{(1-\gamma)(1-\theta)(1-\sigma)}(V-A)+A$, so $\Delta$ depends on the level of lifetime value through $V-A$ and not only on the difference between the two policies. In the stylized model that inverts directly, and every CEV in Section 3.6 is computed that way.

In the quantitative model the same object is read off a curve, because $A$ there is a sum over sixty ages, every health state and every survival path. The benchmark's choices are held fixed, its newborn's expected lifetime value, childhood included, is evaluated at a grid of proportional scalings of consumption, and $\Delta$ is the scaling at which that value equals the counterfactual's, found by interpolating on the curve. Run with no

Table 7: The long run under each policy

| | Housing First | Treatment First | Housing First, $\omega_c = 1$ |
|---|---|---|---|
| *Adults* | | | |
| Drug price, benchmark = 1 | 1.00 | 0.87 | 0.86 |
| Voucher probability $\pi_V$ | 6.75 | 8.14 | 6.75 |
| Drug use | 8.12 | 5.67 | 5.22 |
| Carrying a SUD | 3.46 | 2.39 | 2.21 |
| Homelessness | 0.61 | 0.64 | 0.53 |
| SUD share of the homeless | 19.2 | 21.5 | 11.2 |
| In treatment, of those with a SUD | 25.0 | 29.3 | 25.0 |
| *Children, households aged 26 to 50* | | | |
| Living with a parent who uses drugs | 14.80 | 10.40 | 9.52 |
| Living with a parent who carries a SUD | 5.28 | 3.69 | 3.41 |
| In a shelter | 1.02 | 1.08 | 0.85 |
| *Welfare* | | | |
| CEV of a newborn, percent of lifetime consumption | – | 1.39 | – |
| Share of the newborn's gain that comes in childhood | – | 50 | – |

Notes: percent unless the row says otherwise; each column is a steady state. Housing First is the benchmark. Treatment First screens with a detection rate of 85 percent, and $\pi_V$ holds net government revenue at its benchmark level. The last column is Housing First with $\omega_c = 1$ and $\pi_V$ at its benchmark value. Children are those of households aged 26 to 50, the years a child lives at home. The CEV is against the benchmark, net of the no-policy run's own.

change of policy, the counterfactual's code itself reads 0.027 percent off that curve, and every CEV reported is net of it.

## 6.2 The long run

Table 7 sets the long run under Treatment First beside the benchmark and full altruism. Treatment First cuts drug use by 30 percent and the share of households carrying a SUD by 31 percent, from 3.46 to 2.39 percent of households, and it raises homelessness by 6 percent. Users who lose their vouchers leave the programme, so it costs less at a given $\pi_V$, and $\pi_V$ rises from 6.75 to 8.14 percent, by 21 percent, to hold net government revenue at its benchmark level. Taking users out of the drug market lowers the drug price by 13 percent, which draws some households back and so works against the reform.[14]

The last column keeps Housing First but has each parent weigh its child as itself, $\omega_c = 1$: the drug use parents would choose if they took in full what their use does to their children. Treatment First delivers 85 percent of the fall in drug use that full altruism would bring, 30 percent against 36. The two part ways on homelessness: under full altruism it falls by 13 percent, and the share of children in a shelter falls by 16 percent. $\pi_V$ is held at its benchmark value in this column, so net revenue is not, and it has no CEV, since the weight on the child changes the welfare function itself.

What the reform does to children runs in both directions. The share of children living with a parent who carries a SUD falls from 5.28 to 3.69 percent, by 1.59 percentage points, and the share living with a parent who uses drugs falls from 14.80 to 10.40 percent. The share of children in a shelter rises from 1.02 to 1.08 percent, by 0.06 points. The rise in homelessness is largest among households aged 35 to 44, whose children are of school age: their homelessness rises by 9 percent, against 1 percent at 26 to 34.

A newborn is better off under Treatment First by 1.39 percent of lifetime consumption. About 50 percent of its gain in lifetime utility comes in childhood, from a parent who spends more on the consumption and housing they share and less on the drug. The rest comes in adult life, and only about 21 percent of that from the higher productivity a child draws when its parent does not carry the disorder. The larger part is a higher value of adult life at a given draw: the value of entering the economy at 26 rises at every level of productivity, about 23 times as much at the lowest as at the highest, and the economy a child enters has a cheaper drug and a voucher that is easier to get.

[14] With a drug supply elasticity of 5 rather than 1.46, the price falls by only 11 percent, drug use falls by 45 percent rather than 30, homelessness rises by 2 percent, and the newborn's gain rises to 2.74 percent of lifetime consumption, 74 percent of it in childhood.

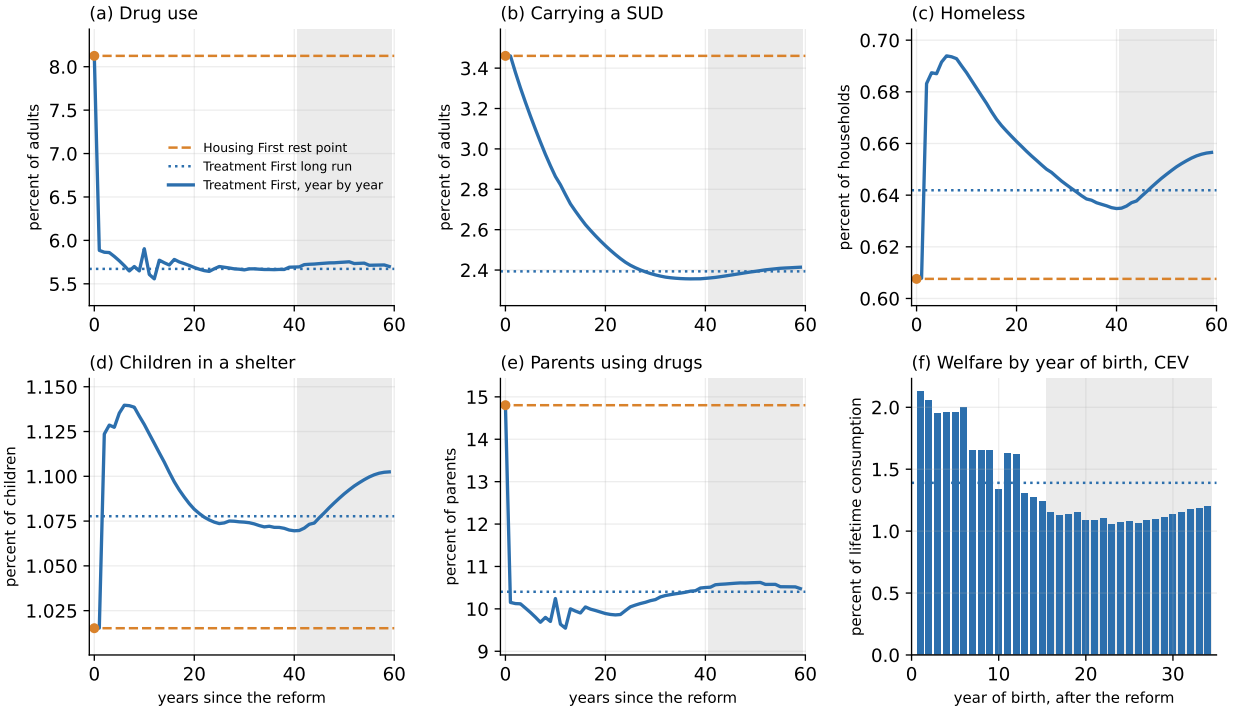



Figure 3: The transition from Housing First to Treatment First, quantitative model

Notes: panels (a) to (e) are by year since the reform, which lands at the start of year 1, with the benchmark as year 0; the dashed orange line is the Housing First benchmark, the dotted blue line the Treatment First long run and the solid blue line the path. Children are those of households aged 26 to 50. Panel (f) is the CEV of each cohort born on the path, against the benchmark, by its year of birth. The path is from a computation that has not yet converged, and from year 41 it drifts away from the steady state it should reach; those years are shaded, and so are the cohorts born from year 16, whose adult lives start in them.

## 6.3 The transition

The transition starts from the benchmark, with the reform landing at the start of the first year, and runs sixty years to the Treatment First steady state; $\pi_V$ and the drug price are solved year by year, so that the drug market clears in every year and the budget in every year after the first, whose spending is set by the vouchers households already hold. The numbers below come from a computation that has not yet converged, and are preliminary. Figure 3 shows the path.

Drug use falls at once: 91 percent of its long-run fall happens in the first year, and the share of parents using drugs falls from 14.80 to 10.15 percent, below its long-run level of 10.40 percent. The share of households carrying a SUD falls slowly, because recovery is slow: half of its long-run fall is done by year 9, and nine tenths by year 21. That stock keeps demand for the drug up, and the drug price stays above its long-run level for 28 years. The elevated price deters young adults who have not yet started: in the first decade the number of drug users without a SUD falls by up to 48 percent, against 29 percent in the long run.

Homelessness overshoots. It rises by 14 percent at its peak, in year 6, against 6 percent in the long run, and then declines slowly toward it. So does the share of children in a shelter, from 1.02 to 1.14 percent in year 6, against 1.08 percent in the long run.

Children born at the reform gain about 2.1 percent of lifetime consumption, against 1.39 percent in the long run, because their parents are the young adults the elevated price keeps from starting, and children born in the first ten years gain 1.83 percent on average. Of the adults alive at the reform, 97 percent are better off under it, but only 14 percent of those who carry a SUD when it lands.

# 7 Discussion

**What the trials can and cannot settle.** The trials measure housing stability over six to twenty four months, and on that margin the model agrees with them: in the first years after the reform Treatment First raises homelessness by up to 14 percent, against 6 percent in the long run, and the share of children in a shelter with it. What decides the ranking moves on a slower clock. Drug use falls at once, but the share of households carrying a SUD takes 21 years to complete nine tenths of its fall, and the gain that puts Treatment First ahead comes to children over their whole lives. A trial of one rule against the other would find Housing First ahead on the outcomes it measures, and would be right about them; it would be silent on the ones that reverse the ranking.

**Which channel carries the result.** About 50 percent of a newborn's gain comes in childhood, from a parent who spends more on what it shares with its child and less on the drug. Of the rest, which comes in adult life, only about 21 percent is the better draw a child inherits from a parent free of the disorder; the rest is a better economy to enter, with a cheaper drug and a voucher that is easier to get. The comparison with full altruism shows what conditionality can and cannot stand in for. If parents weighed their children as themselves, drug use would fall by 36 percent, and Treatment First delivers 85 percent of that fall. But under full altruism homelessness falls by 13 percent, and the share of children in a shelter falls by 16 percent, where Treatment First raises both. Conditionality gets the drug margin roughly right and the housing margin wrong, because it works by taking housing away.

**What would overturn it.** The drug market is where the result is most exposed. Treatment First lowers the drug price, which draws some households back, and the less elastic the supply of the drug the larger that offset. The benchmark takes the elasticity at 1.46, the low end of what the literature uses (Section 2), so it carries a large offset, and with a more elastic supply the newborn's gain rises (footnote 14).

**What the model does not have.** A child is not penalised for growing up in a shelter: homelessness passes from parent to child only through the productivity the child draws. A penalty of that kind would work against Treatment First, but on few children: the share of children in a shelter rises by 0.06 percentage points while the share living with a parent who carries a SUD falls by 1.59. Treatment is a choice at a cost and is never rationed, so the households Treatment First screens out can always enter it. The detection rate of the screen is set rather than estimated. And the transition of Section 6.3 comes from a computation that has not yet converged.

# 8 Conclusion

*Plan for this section, not its content.* Short. The moral hazard the debate argues about is real and is not an inefficiency, so it cannot carry the case for conditionality on its own. The case has to rest on an externality, on the distortions that fund the programme, or on the programme's own distortions, and those are three different claims. This paper prices the first and finds it intergenerational: what a parent's disorder and homelessness cost the child who replaces it. Close on what that implies for a policy debate currently being settled by executive order.

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